

Bloch's conjecture for surfaces with $p_g = 0$

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Abstract

We prove Bloch's conjecture for smooth connected projective complex surfaces with $p_g = 0$: the Albanese homomorphism on integral degree-zero zero-cycles is an isomorphism.

1 Introduction

For a smooth connected projective complex surface S , write

$$p_g(S) = h^0(S, \omega_S), \quad q(S) = h^1(S, \mathcal{O}_S).$$

The Chow group $\mathrm{CH}_0(S)$ is the group of integral zero-cycles modulo rational equivalence, where the relations are generated by divisors of rational functions on curves in S . The degree of a complex point is one, and $\mathrm{CH}_0(S)^0$ denotes the kernel of the degree map. The Albanese homomorphism $\mathrm{alb}_S : \mathrm{CH}_0(S)^0 \rightarrow \mathrm{Alb}(S)(\mathbb{C})$ is surjective. Bloch's conjecture predicts that it is an isomorphism when $p_g(S) = 0$. We prove the conjecture. The main theorem treats the case $q = 0$, when the Albanese variety is a point and the prediction is that any two complex points are rationally equivalent.

Theorem 1.1. *Let S be a smooth connected projective surface over $\mathbb{C}$ with $p_g(S) = q(S) = 0$. Then*

$$\mathrm{deg} : \mathrm{CH}_0(S) \longrightarrow \mathbb{Z}$$

is an isomorphism, where $\mathrm{CH}_0(S)$ has integral coefficients.

Theorem 1.1 imposes no minimality, fundamental-group, or surface-construction hypothesis. Combining it with the classical non-general-type case gives the full statement.

Corollary 1.2 (Bloch's conjecture for surfaces with $p_g = 0$). *Let S be a smooth connected projective surface over $\mathbb{C}$ with $p_g(S) = 0$. Then the Albanese homomorphism*

$$\mathrm{alb}_S : \mathrm{CH}_0(S)^0 \longrightarrow \mathrm{Alb}(S)(\mathbb{C})$$

is an isomorphism of abelian groups, where $\mathrm{CH}_0(S)$ has integral coefficients.

For a surface of general type, $p_g = 0$ forces $q = 0$, so Theorem 1.1 applies. The other cases are the theorem of Bloch–Kas–Lieberman [BKL76]. We give the precise reduction in Section 8.1.

These zero-cycle statements also have consequences for rational Chow motives. For $p_g = q = 0$, every rational Chow class on a power of S or a Hilbert scheme of points is determined by its cohomology class. We state the precise consequences and their motivic antecedents after the proof overview.

History and antecedents

Mumford showed that a nonzero holomorphic two-form prevents the zero-cycles on a surface from admitting a finite-dimensional parametrization modulo rational equivalence [Mum69]. Bloch’s conjecture predicts the converse at the level of the Albanese kernel: if $p_g = 0$, degree and the Albanese image should account for all rational-equivalence classes of zero-cycles [Blo10]. Bloch, Kas, and Lieberman proved this integral assertion for every surface of Kodaira dimension less than two [BKL76, pp. 135–136]. Thus the remaining general-type problem concerns $p_g = q = 0$, where the Albanese variety carries no information and the degree map is predicted to determine a zero-cycle completely.

Several successful approaches use additional geometry of the surface. The group-action method of Inose–Mizukami [IM79] and its extension by Barlow [Bar85] compare zero-cycles with their images on suitable quotient surfaces. Bauer’s treatment of Inoue surfaces with $K^2 = 7$ gives a later application, expressed through relations among the quotient correspondences [Bau14, Theorem 0.2 and Section 1]. Pedrini–Weibel recast involution arguments using the transcendental motive: for a bidouble cover, vanishing of the transcendental motives of the three intermediate quotients forces its own vanishing [PW16, Theorem 2.5]. Their applications include numerical Godeaux and classical Campedelli examples, Keum–Naie, Burniat, and Inoue surfaces, giving examples for each $1 \leq K^2 \leq 8$.

A different route uses families of surfaces and the actions of correspondences on zero-cycles. For a family of regular surfaces whose fibered self-product has a rationally connected smooth projective compactification, Voisin shows that a relative correspondence acting trivially on holomorphic two-forms acts nilpotently on degree-zero zero-cycles. She applies this theorem to Catanese surfaces [Voi14, Theorems 0.3 and 0.6]. Passing to suitable quotients then gives Bloch’s conjecture for Barlow surfaces. This extends the earlier result for Barlow surfaces carrying extra symmetries to the general members of that family. The present argument requires no prescribed family, covering construction, or automorphism of the surface.

Recent preprints propose proofs with overlapping scope. Guletskii’s Theorem A assumes algebraicity of total cohomology and claims vanishing of the Albanese kernel over algebraically closed fields of characteristic zero [Gul25, Theorem A and Section 2.1]. Over $\mathbb{C}$, this cohomological assumption is equivalent to $p_g = q = 0$, by Hodge theory and the Lefschetz $(1, 1)$ theorem, so it covers the full scope of Theorem 1.1. His proposed argument studies a hypersurface section through the diagonal in the square of the surface, together with a nilpotence argument for correspondences. Banerjee proposes a separate proof for minimal general-type surfaces with $p_g = q = 0$ and $K^2 = 9$, using pluricanonical curves and their Jacobians [Ban26, Theorems 1.1 and 2.1]. We cite these as proposed proofs; the argument developed here obtains its Chow relation from virtual localization on a Quot scheme.

The use of the diagonal to control zero-cycles has its antecedents in Bloch–Srinivas [BS83]. Surface Quot obstruction theories and localization relations were developed by Marian–Oprea–Pandharipande [MOP17], with subsequent virtual intersection theory in [OP21] and its K -theoretic counterpart in [AJLOP21]. Ciocan-Fontanine–Kapranov constructed derived Quot schemes [CFK01]. We use the derived moduli and determinant constructions of Toën–Vaquié and Schürg–Toën–Vezzosi [TV07, STV15] to impose the determinant constraint. The Chow calculation adapts the nested-Hilbert-scheme recursion of Ellingsrud–Göttsche–Lehn [EGL01] by leaving two surface factors unintegrated throughout. This is the step that produces a correspondence identity rather than only a numerical intersection formula.

The mechanism of the proof

The rational cohomology of a surface with $p_g = q = 0$ is generated by algebraic classes, but cohomological identities alone do not settle rational equivalence. After the classical and birational reductions, we work on a minimal surface of general type or a point blowup X of that surface. The objective is a relation

$$0 = c[\Delta_X] + \Gamma \quad \text{in } \mathrm{CH}^2(X \times X)_{\mathbb{Q}}, \quad (1.1)$$

where $c \neq 0$ and Γ is a sum of external products of cycles. The diagonal acts as the identity on zero-cycles, whereas every external product of total codimension two kills degree-zero zero-cycles. Thus the relation gives $\mathrm{CH}_0(X)^0 \otimes \mathbb{Q} = 0$. Constructing this Chow relation and detecting its coefficient are the two main stages.

For line bundles L_1, L_2, D , consider rank-zero quotients of $L_1 \oplus L_2$ whose determinant has Picard class D . Opposite torus weights on the summands give a localization formula. We construct its obstruction theory and the compatible fixed theory using a derived determinant fiber normalized at a point of X (Sections 3 and 4). A fixed quotient splits into a quotient of each line bundle, each described by an effective divisor and a zero-dimensional subscheme. The divisor line bundles T, H satisfy $T + H = D$; on an irregular surface this determinant condition also relates their Picard parameters.

Two second-Chern-character insertions of the universal quotient leave a class on $X \times X$. We choose the virtual dimension so that equivariant degree forces its codimension-two part to vanish. To calculate it in Chow, we retain both output copies of X throughout the Hilbert-scheme recursion of Ellingsrud–Göttsche–Lehn [EGL01]. The recursion replaces the point incidence classes by diagonals identifying surface coordinates. Terms joining both outputs contribute a multiple of $[\Delta_X]$; terms with separated outputs contribute external products. For each fixed fiber expression, this procedure chooses its diagonal coefficient from the intersection numbers of its line bundles and the Chern numbers of the surface. The total c is the weighted sum over the fixed contributions that occur (Section 5). The rule does not require a diagonal decomposition on an individual surface to be unique.

For a minimal general-type surface with $p_g = q = 0$ and canonical class K , one has $b_2 = 10 - K^2$. When $K^2 \leq 8$, its lattice has a direction orthogonal to the canonical class. A lattice construction and at most three point blowups provide a class α with $\alpha^2 < 0$ and Quot data with the following property. For each contributing fixed pair with $\alpha H = 0$, pairing with $\alpha \boxtimes \alpha$ kills its external terms. Every remaining contributing pair has zero point lengths, so its chosen diagonal coefficient is zero, while its pairing is $(\alpha H)^2$. Consequently the full relation gives

$$0 = c\alpha^2 + \sum_{\alpha H \neq 0} (\alpha H)^2.$$

The sum runs over a finite set of effective fixed pairs, and a separate effectivity argument shows that at least one occurs. Since $\alpha^2 < 0$, this proves $c > 0$ (Section 6).

When $K^2 = 9$, the minimal surface has no such orthogonal direction. We choose $D = 3K$ and arrange that only the numerical divisor pairs $(3K, 0)$ and $(2K, K)$ can contribute to the diagonal coefficient. Write c_{out} and c_{mid} for their respective contributions, including both orders and all admissible point lengths. They can be compared on the Cartwright–Steger surface, which has the same Chern numbers but $p_g = q = 1$. The real part of its holomorphic two-form has nonzero square and is orthogonal to all divisor classes, so pairing detects the diagonal coefficient. The divisor cosection of Chang–Kiem [CK13], with Kiem–Li localization [KL13], reduces its fixed contributions to fibers over the empty and canonical divisors. These have virtual multiplicities 1 and $-s$ with $s > 0$. After restricting to those fibers, the universal rule compares the same two coefficients on

both surfaces and gives $c_{\text{out}} = sc_{\text{mid}}$. A length-two Hilbert-scheme calculation gives $c_{\text{mid}} = 12$. On the original surface the two coefficients occur with nonnegative multiplicities, and the outer type occurs once, so its total coefficient is positive (Section 7).

Finally, rational vanishing makes each integral degree-zero class torsion. Roitman's torsion theorem, together with $q = 0$, makes it zero [Blo79, Roi80].

The technical constructions used to obtain (1.1) are recorded separately. They include compatible obstruction theories for the based determinant constraint (Propositions 3.4 and 3.6), an excess comparison for rank-one Quot schemes (Lemma 4.6), and a Chow-valued universality statement with two marked surface factors. The latter applies to the tautological pushforwards specified in Proposition 5.3, independently of the numerical choices made for the zero-cycle application.

Figure 1 records how the numerical branches feed into the same diagonal criterion.

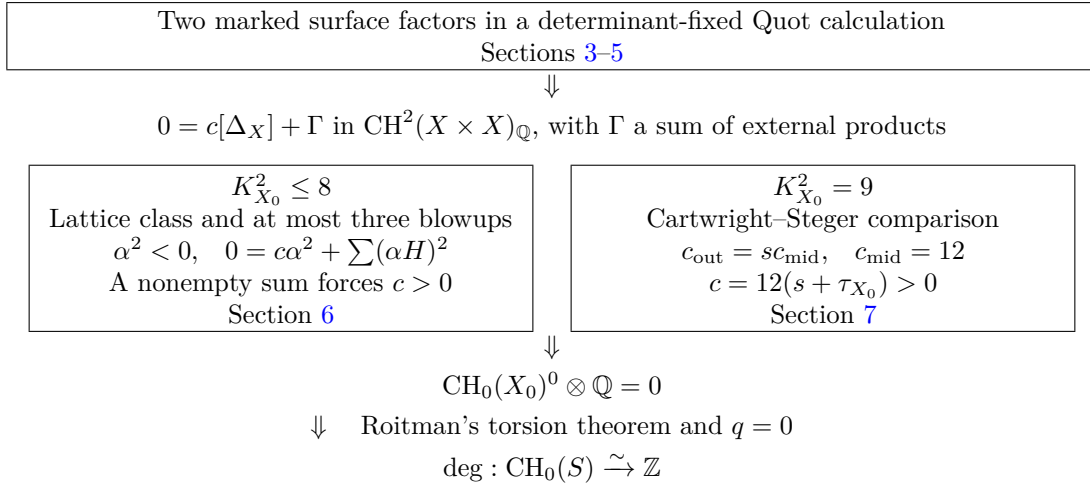


Figure 1: The general-type proof, after passage to the minimal model X_0 of S . Each branch chooses Quot data and proves that the resulting diagonal coefficient is nonzero. In the left branch, the displayed squares come from the external-product terms with $\alpha H \neq 0$; those terms have zero diagonal coefficient. In the right branch, the outer and middle coefficients are defined by the same numerical rule on the target and auxiliary surfaces; $-s$ is the canonical-divisor virtual coefficient on the auxiliary surface, with $s > 0$, and τ_{X_0} counts the nontrivial torsion line bundles. Birational invariance returns the conclusion to S .

Motivic consequences

The full zero-cycle statement also has consequences for rational Chow motives. Write $h(Y)_{\mathbb{Q}}$ for the rational Chow motive of a smooth projective complex variety Y , and $t_2(S)$ for the transcendental degree-two summand defined by Kahn–Murre–Pedrini. Let $\mathbb{L}$ denote the effective Lefschetz motive, whose Betti realization is $\mathbb{Q}(-1)$ in degree two. A rational Chow motive is *pure Tate* if it is a finite direct sum of powers of $\mathbb{L}$.

Corollary 1.3 (Motivic consequences). *Let S be a smooth connected projective surface over $\mathbb{C}$ with $p_g(S) = 0$. Then $h(S)_{\mathbb{Q}}$ is finite-dimensional in the sense of Kimura and $t_2(S) = 0$.*

If, in addition, $q(S) = 0$, then

$$h(S)_{\mathbb{Q}} \simeq \mathbf{1} \oplus \mathbb{L}^{\oplus b_2(S)} \oplus \mathbb{L}^2.$$

In this case, for every $m, n \geq 0$, the rational Chow motives of the power S^m and the Hilbert scheme $S^{[n]}$ of length- n zero-dimensional subschemes are pure Tate, with $S^0 = S^{[0]} = \text{Spec } \mathbb{C}$. For $Y = S^m$ or $Y = S^{[n]}$, the rational cycle-class maps satisfy

$$\text{cl}_Y^i : \text{CH}^i(Y)_{\mathbb{Q}} \xrightarrow{\sim} H^{2i}(Y, \mathbb{Q}), \quad H^{2i+1}(Y, \mathbb{Q}) = 0 \quad \text{for every } i \geq 0.$$

These are consequences of established motivic criteria. In the framework of Kimura's finite-dimensional motives [Kim05], Guletskiĭ–Pedrini proved the equivalence with Bloch's conjecture for surfaces with $p_g = 0$ and recorded the resulting Tate decomposition when $q = 0$ [GuP03, Theorem 27 and Remark 28]. The surface decomposition of Kahn–Murre–Pedrini identifies vanishing of the Albanese kernel with $t_2(S) = 0$ [KMP07]; the Hilbert-scheme assertion then uses the motive formula of de Cataldo–Migliorini [dCM02]. In particular, every rational Chow class on the stated powers and Hilbert schemes is determined by its cohomology class. This conclusion uses the Tate decomposition, not finite-dimensionality alone. Section 8.2 gives the proof.

Section 2 first passes to minimal surfaces of general type. Section 3 constructs the determinant-fixed theory and its fixed part; Section 4 computes the resulting virtual classes and the divisor cosection. Section 5 proves the marked Chow relation and its action on zero-cycles. Sections 6 and 7 prove the two nonvanishing statements. Section 8.1 gives the integral conclusion, and Section 8.2 derives the motivic consequences.

Conventions

All schemes and cohomology groups are over $\mathbb{C}$. The definitions of $\text{CH}_0(S)$ and alb_S above, Theorem 1.1, Corollary 1.2 and its proof, Lemma 2.1, and Section 8.1 use integral Chow groups. Elsewhere Chow groups have rational coefficients; in particular, all Chow motives in Section 8.2 have rational coefficients. We use cohomological indexing for complexes. Line bundles are written additively; in intersections the same symbol denotes their first Chern class. For a perfect complex, its Euler class with a nonzero torus weight is interpreted in localized equivariant Chow theory. Statements about virtual classes always use the specified perfect obstruction theories. A comparison of their classes in K -theory is used to compute Chern classes only after the relevant obstruction bundle or compatible map of theories has been identified.

2 Birational reductions

We reduce the zero-cycle theorem to minimal surfaces of general type and record the numerical range that determines the two branches of the proof. We also retain the freedom to blow up points, which will be used in Section 6.

Lemma 2.1. *Let $\pi : \tilde{X} \rightarrow X$ be the blowup of a smooth projective surface at a point. Then p_g and q are unchanged, and*

$$\pi_* : \text{CH}_0(\tilde{X}) \xrightarrow{\sim} \text{CH}_0(X)$$

is a degree-preserving isomorphism with integral coefficients. Consequently these assertions are invariant under birational maps between smooth projective surfaces.

Proof. The identities $\pi_* \mathcal{O}_{\tilde{X}} = \mathcal{O}_X$ and $R^1 \pi_* \mathcal{O}_{\tilde{X}} = 0$ give invariance of q . Pullback identifies holomorphic two-forms: a form on $\tilde{X}$ descends away from the blown-up point and extends across that point on the smooth surface. This proves invariance of p_g . The integral blowup formula for Chow groups [Ful98, Proposition 6.7(b), (e)] gives, in codimension two,

$$\text{CH}^2(\tilde{X}) \simeq \text{CH}^2(X) \oplus \text{CH}^1(\text{Spec } \mathbb{C}) = \text{CH}^2(X).$$

The first summand is pulled back from X , and pushforward is its inverse. Proper pushforward preserves degree. Resolving a birational map of smooth projective surfaces by point blowups gives the last assertion. See also [Blo79, Section 4, first reduction]. $\square$

Proposition 2.2. *To prove Theorem 1.1 with rational coefficients, it is enough to prove it for smooth minimal surfaces X_0 of general type with $p_g = q = 0$. For such a surface, putting $k_0 = K_{X_0}^2$, one has*

$$1 \leq k_0 \leq 9, \quad \chi(\mathcal{O}_{X_0}) = 1, \quad \int_{X_0} c_2(X_0) = 12 - k_0, \quad b_2(X_0) = 10 - k_0. \quad (2.1)$$

It is also enough to prove the assertion on any chosen sequence of point blowups of each such X_0 .

Proof. For Kodaira dimension less than two, the Theorem of Bloch, Kas, and Lieberman gives zero Albanese kernel [BKL76, p. 135]. Since $q = 0$, this is the full degree-zero Chow group. For a surface of general type, pass to a smooth minimal model and apply Lemma 2.1.

On the minimal model, K_{X_0} is nef and big, so $k_0 > 0$. The vanishing of p_g and q gives $\chi(\mathcal{O}_{X_0}) = 1$. Noether's formula gives $\int c_2 = 12 - k_0$. Hodge theory and Poincaré duality give $b_1 = b_3 = 0$, and hence $\int c_2 = 2 + b_2$. Therefore $b_2 = 10 - k_0$. An ample divisor has a nonzero cohomology class, so $b_2 \geq 1$, proving $k_0 \leq 9$. The final assertion is another application of Lemma 2.1. $\square$

3 The determinant-fixed Quot obstruction theory

We make the virtual construction in two settings:

- (i) X is a smooth connected projective complex surface with $p_g = q = 0$;
- (ii) X is a smooth connected projective complex surface with $p_g = q = 1$, $K = K_X$ ample, and $D = 3K$.

Write $K = K_X$. In both settings $\chi(\mathcal{O}_X) = 1$. The second setting is used only to determine universal coefficients in Section 7. No positivity assumption on K is imposed in setting (i). All virtual classes and equivariant Chow groups in this Section have rational coefficients.

Our goal is an equivariant perfect obstruction theory on the determinant-fixed Quot scheme, together with its actual restriction to the fixed locus. Surface Quot obstruction theories and localization relations for a trivial target bundle were developed by Marian, Oprea, and Pandharipande [MOP17, Section 1]; subsequent intersection-theoretic and K -theoretic developments appear in [OP21, AJLOP21]. Here the based determinant fiber must also work on the irregular auxiliary surface. We construct that fiber as a derived moduli problem so that its obstruction morphism, and not only its virtual tangent class, is specified.

Fix line bundles L_1, L_2, D , and put $V = L_1 \oplus L_2$ and $A = L_1 - L_2$. For an integer n , let B be the scheme of quotients

$$0 \longrightarrow S' \xrightarrow{i} V \longrightarrow Q \longrightarrow 0, \quad (3.1)$$

$$\mathrm{rk} Q = 0, \quad \det Q = D, \quad \int_X \mathrm{ch}_2(Q) = n + L_1 D - \frac{D^2}{2}.$$

Here the determinant condition is equality in the Picard scheme. Choose a polarization on X . The stated Chern data determine a Hilbert polynomial, and B is the determinant fiber in the corresponding projective Quot scheme. In particular, B is projective. The group $G = \mathbb{G}_m$ acts on V with characters $+1, -1$ on its two summands, and hence acts on B . We write $t = c_1^G(\mathbb{C}_1)$ for the

equivariant parameter. Later we specialize t to $1/2$, so the evaluated weights of the two summands are $w_1 = 1/2$ and $w_2 = -1/2$.

Our complexes use cohomological grading: a sheaf $F[-1]$ lies in degree one. Pullbacks and tensor products of complexes are derived unless specified otherwise. If Y is a parameter scheme and $\pi : X \times Y \rightarrow Y$ is projection, we abbreviate

$$R\mathrm{Hom}(F, G) = R\pi_* R\mathcal{H}om(F, G).$$

Thus Hom complexes in families are complexes on the parameter scheme. The notation $R\Gamma(F)$ for an object on X denotes its derived sections, pulled back as a constant complex when parameters are present.

3.1 Derived arrows and classical families

Ciocan-Fontanine and Kapranov constructed a derived Quot scheme whose truncation is the classical Quot scheme and whose tangent cohomology is given by Ext groups from the kernel to the quotient [CFK01, Theorem 4.3.2]. Here we use a perfect-arrow moduli description to retain the maps needed for the based determinant fiber.

We use the derived moduli stack of perfect complexes and its morphism stack as constructed in [TV07, Theorem 3.6, Definition 3.18, and Corollaries 3.17, 3.19, and 3.29]. For smooth proper X these stacks are locally geometric and locally of finite presentation. The morphism stack with target fixed to V also has a direct description: over a perfect source F its fiber is the derived linear mapping stack associated to $R\mathrm{Hom}(F, V)$. This complex is perfect, and its formation commutes with derived base change, because X is smooth and proper.

Lemma 3.1 (The open Quot enhancement). *The fixed-target derived arrow stack has an open substack $\mathfrak{U}$ whose truncation U is the ordinary Quot scheme with the chosen rank and Hilbert polynomial, before fixing determinant. On U , its tangent complex is naturally*

$$\begin{aligned} T_{\mathfrak{U}}|_U &\simeq \mathrm{Cone}(R\mathrm{Hom}(S', S') \xrightarrow{\mathrm{id}} R\mathrm{Hom}(S', V)) \\ &\simeq R\mathrm{Hom}(S', Q). \end{aligned} \tag{3.2}$$

It is perfect of Tor-amplitude $[0, 1]$. These identifications are compatible with forgetting the source or target, with direct sums of arrows, and with base change. The same assertions hold for a fixed vector-bundle target of any rank and torsion quotients.

Proof. First take the locus where both the source and its cone are sheaves in degree zero on classical fibers, with the prescribed Hilbert polynomials. We explain the family conditions in this description. For a perfect complex on $X \times Y$, the locus where it has relative Tor-amplitude zero over Y is open. Locally one represents the complex by vector bundles. At a fiber on which it is exact away from degree zero, the positive end can be split off. At the negative end, fiberwise injectivity between modules flat over the base implies injectivity with base-flat cokernel, by the local criterion of flatness. Successively replacing the negative end by these cokernels gives a sheaf flat over the base near that fiber. Openness of flatness gives an open neighborhood in $X \times Y$; properness of $X \times Y \rightarrow Y$ then gives the required open neighborhood in Y . Applying this to the source and cone proves the asserted openness.

Conversely, let $V_Y \rightarrowtail Q$ be a flat quotient family over a noetherian base Y . Its kernel S' is flat over Y . Both Q and S' are perfect on $X \times Y$, even if Y is singular. Indeed, resolve either sheaf locally by finite-rank free modules. Every successive syzygy remains flat over Y . On a surface fiber the local rings are regular of dimension at most two, so after two steps the syzygy is locally free

on that fiber. A finitely presented, base-flat module with locally free fibers is locally free near the point, again by the local flatness criterion. This truncates the resolution. This argument suffices over noetherian bases, and finite presentation extends the moduli identification to arbitrary classical test schemes. Flat approximation and the perfection criteria in [Stacks, Tags 02JO, 0DHY, and 068X] also give this last passage: a finitely presented base-flat module is pseudo-coherent, and the regular surface fibers give absolute Tor-amplitude contained in $[-2, 0]$.

On our open locus the cone sequence is consequently an exact sequence of sheaves with flat quotient. An automorphism a of its source satisfying $i \circ a = i$ is the identity, since i is injective. Negative Ext groups between these sheaves vanish as well, so no higher automorphisms remain in the classical moduli problem. The truncation is therefore the Quot scheme, including its functor on nonreduced bases.

Here is a tangent calculation that also records the relevant maps. Allow both vertices of an arrow $F \rightarrow G$ to vary. Its derived endomorphism complex is

$$\mathrm{fib} \left\{ R\mathrm{Hom}(F, F) \oplus R\mathrm{Hom}(G, G) \xrightarrow{(a,b) \mapsto fa - bf} R\mathrm{Hom}(F, G) \right\}.$$

One can compute this complex with the two-term resolution of a one-arrow diagram by diagrams induced from its vertices. More explicitly, if $P_1 = (\mathcal{O}_X \xrightarrow{\mathrm{id}} \mathcal{O}_X)$ and $P_2 = (0 \rightarrow \mathcal{O}_X)$, the resolution has terms $P_2 \otimes F$ and $(P_1 \otimes F) \oplus (P_2 \otimes G)$. Its map at the target vertex sends s to $(s, -f(s))$. Thus perfect vertices give a perfect diagram. The one-arrow path algebra is finite dimensional and has this finite projective resolution; the usual perfect-object tangent formula applies to its diagrams on X . The tangent of their moduli is the displayed endomorphism complex shifted by [1].

Fixing the target takes the tangent fiber over $R\mathrm{Hom}(V, V)[1]$. The result is the first cone in (3.2); applying $R\mathrm{Hom}(S', -)$ to (3.1) gives the second identification. In particular, forgetting the arrow and remembering its source has differential the boundary

$$R\mathrm{Hom}(S', Q) \longrightarrow R\mathrm{Hom}(S', S')[1]. \quad (3.3)$$

These are identifications of perfect complexes in families. For completeness, perfection and functoriality can also be checked before fixing a vertex by taking pointed loops. The relative loop stack at a perfect family is its self-equivalence stack, open in the linear mapping stack of its perfect endomorphism complex. Its cotangent at the identity is the dual endomorphism complex. The cotangent triangle for the loop fiber product identifies this with the pulled-back moduli cotangent shifted by [1]. Forgetting vertices and taking direct sums act on these loop tangents by projection and block-diagonal sum. Taking the fixed-target fiber preserves these identifications. This proves the claimed functoriality, including the specified boundary.

Finally, at a geometric point S' is torsion-free and Q is torsion. Surface Serre duality gives

$$\mathrm{Ext}^2(S', Q)^\vee = \mathrm{Hom}(Q, S'(K)) = 0.$$

Negative Ext groups vanish for sheaves; groups above degree two vanish on a smooth surface. Consequently the derived tangent has geometric fibers in degrees $[0, 1]$. For a perfect complex the geometric-fiber criterion is precisely the criterion for Tor-amplitude in this range. $\square$

We will use the following comparison in both the absolute and relative constructions. We include the infinitesimal argument to specify which cotangent morphism supplies the obstruction theory.

Lemma 3.2 (Truncation comparison). *Let $\mathfrak{Z}$ be a locally geometric derived stack, locally of finite presentation, whose classical truncation $j : Z \rightarrow \mathfrak{Z}$ is a scheme. Suppose $E_Z = j^* L_{\mathfrak{Z}}$ is perfect of Tor-amplitude $[-1, 0]$. The canonical map*

$$\phi_Z : E_Z \longrightarrow L_Z$$

has cone in $D^{\leq -2}(Z)$, and is a perfect obstruction theory. For a map of such enhancements, the induced cotangent map commutes with the corresponding maps ϕ_Z .

Proof. The cotangent map is induced by j , so its functoriality is that of cotangent complexes. The comparison statement is [STV15, Proposition 1.2 and Corollary 1.4]; the following argument gives it directly in the present scheme-truncation situation. Work on an affine open $x : \text{Spec } R \rightarrow Z$, and write $E = x^* E_Z$. For every ordinary R -module N , split square-zero extensions and agreement of Z and $\mathfrak{Z}$ on classical rings give an isomorphism

$$\text{Hom}(L_R, N) \xrightarrow{\sim} \text{Hom}(E, N).$$

There is also an injection

$$\text{Hom}(L_R, N[1]) \longrightarrow \text{Hom}(E, N[1]).$$

Indeed, let a derivation class $\delta : R \rightarrow R \oplus N[1]$ map to zero on the right. The two points of $\mathfrak{Z}(R \oplus N[1])$ obtained from x by δ and by the zero derivation are then homotopic, over x . Infinitesimal cartesianness of a locally geometric derived stack produces a lift of x over the homotopy pullback

$$R' = R \times_{R \oplus N[1]}^h R,$$

whose two maps are δ and zero. This is the classical square-zero extension of R by N classified by δ . The lift is a classical point of Z and factors through the same open $\text{Spec } R$; it therefore gives a splitting of $R' \rightarrow R$. The extension class δ is zero, proving injectivity. Put $C = \text{Cone}(E \rightarrow L_R)$. The two tests, together with connectivity, give $\text{Hom}(C, N) = \text{Hom}(C, N[1]) = 0$ for every N . Since $C \in D^{\leq 0}$, the first equality implies $h^0(C) = 0$, and the second then implies $h^{-1}(C) = 0$. Thus $C \in D^{\leq -2}$. Together with the given perfect amplitude, this is the definition of a perfect obstruction theory in [BF97, Definitions 4.4 and 5.1]. $\square$

3.2 The based determinant and its amplitude

The unrestricted Quot enhancement is now available. To impose the determinant condition without introducing scalar automorphisms, we first normalize line bundles at one point and then take a derived fiber. The remaining amplitude check concerns its obstruction map.

Choose $o \in X$. Let $\mathfrak{P}^o$ be the derived Picard stack of line bundles on X trivialized at o : it is the fiber, over the trivial line, of evaluation at o from the derived line-bundle stack to $B\mathbb{G}_m$. Its truncation is the Picard scheme $\text{Pic}(X)$. Indeed, every line M on $X \times Y$ has the normalized form

$$\nu(M) = M \otimes \pi^*(M|_{\{o\} \times Y})^{-1},$$

with its canonical trivialization at o . Two lines with the same Picard class differ by a line from the base, which disappears on normalization. Moreover $\pi_* \mathcal{O}_{X \times Y} = \mathcal{O}_Y$, so every line automorphism is a unit from the base; one preserving the trivialization is the identity. These observations identify the truncation as a functor, not only on complex points.

The evaluation tangent is $R\Gamma(\mathcal{O}_X)[1] \rightarrow \mathbb{C}[1]$. Constant functions split it, and hence

$$T_{\mathfrak{P}^o} = R\Gamma(\mathcal{I}_o)[1], \quad R\Gamma(\mathcal{O}_X) = \mathbb{C} \oplus R\Gamma(\mathcal{I}_o). \quad (3.4)$$

In particular its cohomology lies in degrees zero and one, with groups $H^1(\mathcal{O}_X)$ and $H^2(\mathcal{O}_X)$ respectively.

The derived determinant morphism of [STV15, Section 3.1, Proposition 3.2], followed by ν , gives a morphism $\mathfrak{U} \rightarrow \mathfrak{P}^o$. Set

$$\mathfrak{B} = \mathfrak{U} \times_{\mathfrak{P}^o}^h \{\nu(L_1 + L_2 - D)\}. \quad (3.5)$$

Its truncation is B : the fiber fixes the Picard class of $\det S'$ and hence of $\det Q$. It makes no choice of trivialization of the unnormalized determinant of the family at o .

The determinant differential is trace at geometric points [STV15, Proposition 3.2]. We need this identification on the universal complexes over classical parameter schemes, including nonreduced ones, and justify that family statement here. Let Y be such a quotient parameter scheme. The perfect source on $X \times Y$ classifies a map to the derived stack of perfect complexes. Pull back the differential of its determinant morphism to the line-bundle stack $B\mathbb{G}_m$. The perfect-object tangent formula gives an internal map

$$R\mathcal{H}om(S', S')[1] \rightarrow \mathcal{O}_{X \times Y}[1].$$

The determinant on complexes on X is obtained by applying this morphism to families on X ; its differential is the derived pushforward of this internal map.

Perfect trace duality identifies the internal map with an endomorphism a of the sheaf S' . On the locus where S' is locally free, the ordinary determinant differential gives $a = \text{id}_{S'}$. To extend this equality over Y , let Z be the closed non-locally-free locus of S' . Base flatness and the fiber criterion for local freeness show that Z has finite fibers over Y . Near a point of Z , choose f in its ideal whose image in the regular local surface fiber is a nonzero element. That image is a nonzerodivisor; the local flatness criterion makes f a nonzerodivisor on the total space as well. The image of $a - \text{id}_{S'}$, viewed in V_Y , is coherent and supported on Z , so locally a power of f annihilates it. Since V_Y is locally free, this image is zero. Thus $a = \text{id}_{S'}$ everywhere, proving the trace identity as a map of complexes over Y . This argument applies to the universal noetherian parameter schemes used below; naturality and derived pullback preserve the identity on their base changes.

The tangent map defining (3.5) is consequently the composition of (3.3), trace, and the projection in (3.4). We denote this composition by $d\det^o$. The two complexes in this tangent fiber have amplitude $[0, 1]$. Their long exact sequence shows that its only possible extra group is

$$H^2(T_{\mathfrak{B}}|_b) = \text{coker}\left\{\text{Ext}^1(S', Q) \rightarrow H^2(\mathcal{O}_X)\right\}$$

at a geometric point b . The following rank bound makes this group vanish.

Lemma 3.3 (Surjectivity of the determinant obstruction map). *Let $0 \rightarrow F \rightarrow V_0 \rightarrow Q_0 \rightarrow 0$ be an exact sequence on X , where V_0 is a vector bundle of rank r and Q_0 is torsion. The map*

$$\text{Ext}^1(F, Q_0) \rightarrow \text{Ext}^2(F, F) \xrightarrow{\text{tr}} H^2(\mathcal{O}_X)$$

is surjective if $p_g = 0$. It is also surjective if K is ample and $Kc_1(Q_0) > rK^2$.

Proof. There is nothing to prove when $H^2(\mathcal{O}_X) = 0$. Otherwise Serre duality and the exact sequence show that a section $\omega \in H^0(K)$ annihilates the image precisely when $\omega \text{id}_F \in \text{Hom}(F, F(K))$ lifts to a map

$$a : V_0 \rightarrow F(K), \quad a|_F = \omega \text{id}_F.$$

Writing $i : F \rightarrow V_0$, the maps $i(K)a$ and ωid_{V_0} agree where $F = V_0$, a dense open because Q_0 has rank zero. They are maps between vector bundles, so they agree everywhere. Composing with the quotient to $Q_0(K)$ gives $\omega Q_0 = 0$.

Suppose now that $\omega \neq 0$, and write its effective divisor as $C = (\omega)$. The quotient Q_0 factors through $V_0|_C$. At the generic point of any integral curve $C_j \subset X$ the local ring is a DVR. If ω has order a_j there, a quotient of $(\mathcal{O}_{X,C_j}/(\omega))^r$ has length at most ra_j . The divisorial cycle of a torsion sheaf is the sum of these generic lengths, and represents its first Chern class: the elementary-divisor decomposition over a DVR identifies this length with the order of the determinant. Therefore

$$[Q_0]_{\text{div}} \leq rC \quad \text{componentwise}, \quad Kc_1(Q_0) \leq rK^2.$$

The latter inequality uses ampleness of K . It contradicts the hypothesis. The argument allows embedded points and nonreduced curve support, since it uses only generic lengths. Thus no nonzero ω annihilates the image, proving surjectivity. $\square$

Proposition 3.4 (The determinant-fixed Quot theory). *In either setting (i) or (ii), the enhancement $\mathfrak{B}$ induces a perfect obstruction theory $E_B \rightarrow L_B$, with virtual tangent*

$$E_B^\vee \simeq \text{fib} \left\{ R\text{Hom}(S', Q) \xrightarrow{d \det^o} R\Gamma(\mathcal{I}_o)[1] \otimes \mathcal{O}_B \right\}. \quad (3.6)$$

It is G -equivariant. Its virtual dimension is

$$v = 2n + (A - K)D. \quad (3.7)$$

Proof. The cotangent triangle of the derived fiber gives (3.6) as an identity of perfect complexes. Both its input and target have Tor-amplitude $[0, 1]$; the input has this amplitude by Lemma 3.1. Their fiber has no negative cohomology and has no cohomology above degree two. Its degree-two cohomology vanishes exactly when

$$\text{Ext}^1(S', Q) \longrightarrow H^2(\mathcal{O}_X)$$

is surjective at every geometric point. In setting (i) its target is zero. In setting (ii), Lemma 3.3 applies with $r = 2$, since $KD = 3K^2 > 2K^2$. The geometric-fiber criterion for perfect complexes now gives amplitude $[0, 1]$. Lemma 3.2 therefore supplies the specified obstruction morphism.

Postcomposition with the given G -action on V defines an action on the arrow stack, leaving the source determinant map invariant. It thus acts on the fiber (3.5), and naturality of the cotangent map makes its obstruction theory equivariant. Equivalently, one can construct the truncation map on the quotient stacks relative to BG , obtaining exactly this equivariant complex and morphism.

The Euler characteristic of the determinant target is $q - p_g = 0$. If $q_2 = \int_X \text{ch}_2(Q)$, Riemann–Roch gives

$$\chi(V, Q) = 2q_2 - (K + L_1 + L_2)D, \quad \chi(Q, Q) = -D^2.$$

For the second identity, the only degree-four term in $\text{ch}(Q^\vee) \text{ch}(Q) \text{td}(X)$ is $-D^2$, since Q has rank zero; here $Q^\vee$ is the derived dual. Consequently

$$\begin{aligned} v &= \chi(S', Q) - (q - p_g) = \chi(V, Q) - \chi(Q, Q) \\ &= 2 \left(n + L_1 D - \frac{D^2}{2} \right) - (K + L_1 + L_2)D + D^2 = 2n + (A - K)D. \end{aligned}$$

$\square$

3.3 Equivariant resolutions and the fixed theory

The determinant-fixed obstruction theory is constructed. To apply virtual localization, we still need global equivariant presentations and a comparison identifying the fixed obstruction morphism itself.

Lemma 3.5 (Equivariant global presentations). *The scheme B has a G -equivariant closed embedding in a smooth projective scheme. Its obstruction theory has a global presentation by two G -equivariant vector bundles. The morphism to the truncated cotangent complex can be represented by a morphism of two-term complexes for this embedding.*

Proof. The Grassmannian construction of the projective Quot scheme is equivariant for the action on V . The determinant fiber is closed and invariant, so restriction and a Plücker embedding give an equivariant embedding $B \hookrightarrow M$ with M smooth projective. In particular B has a linearized ample bundle $\mathcal{O}_B(1)$.

Here is the global resolution argument for $E = E_B$. For m sufficiently large, $h^0(E)(m)$ is generated by sections and $H^a(h^{-1}(E)(m)) = 0$ for every $a > 0$. The hypercohomology sequence then lifts its sections to $\mathbb{H}^0(E(m))$. These are finite-dimensional torus representations; linear reductivity allows an equivariant choice of lifts. Evaluation gives an equivariant bundle $V_0 = \mathcal{O}_B(-m) \otimes W_0$ and a map $V_0 \rightarrow E$ surjective on h^0 . Its derived fiber V_1 is perfect. At every geometric point b , the same map is surjective on h^0 , since $h^0(E \otimes^L k(b)) = h^0(E) \otimes k(b)$. Its fiber has cohomology only in degree zero. Thus V_1 is a vector bundle, and $E \simeq [V_1 \rightarrow V_0]$ in degrees $[-1, 0]$.

Let J be the conormal sheaf of $B \hookrightarrow M$, and set $\Omega = \Omega_M|_B$. The truncated cotangent complex $\tau_{\geq -1}L_B$ is represented by $C = [J \rightarrow \Omega]$ in degrees $[-1, 0]$. Increase m above so that $\text{Ext}^1(V_0, J) = 0$ as well; this is Serre vanishing for $J(m)$. The triangle $J \rightarrow \Omega \rightarrow C \rightarrow J[1]$ then lifts $V_0 \rightarrow E \rightarrow C$ to $V_0 \rightarrow \Omega$, equivariantly. The map $E \rightarrow C$ includes a nullhomotopy on its restriction from V_1 . Together with the chosen lift, that nullhomotopy gives a map $V_1 \rightarrow J = \text{fib}(\Omega \rightarrow C)$. These are ordinary sheaf maps and give a commuting square

$$\begin{array}{ccc} V_1 & \longrightarrow & V_0 \\ \downarrow & & \downarrow \\ J & \longrightarrow & \Omega \end{array}$$

representing the original map to C . All choices can be made in the equivariant category, since taking torus invariants is exact. $\square$

On the classical fixed locus, an invariant subsheaf of $L_1 \oplus L_2$ decomposes as the direct sum of its two character subsheaves. This holds over every classical base with trivial G -action: the coaction decomposes a module into its weight modules, and the inclusion in V admits only the two indicated weights. The kernels and quotients in these summands are flat because they are direct summands of flat families. Thus a fixed arrow is precisely a pair

$$S'_1 \longrightarrow L_1, \quad S'_2 \longrightarrow L_2,$$

of rank-one kernel arrows with torsion quotients Q_1, Q_2 , subject to the total data and determinant constraint. There are finitely many discrete summand Hilbert polynomials on the fixed scheme: they are locally constant, and the fixed scheme is of finite type.

Write $\mathfrak{U}_i$ for the unrestricted derived arrow Quot of Lemma 3.1 with target L_i . For each choice of the discrete data, let

$$\mathfrak{W} = (\mathfrak{U}_1 \times \mathfrak{U}_2) \times_{\mathfrak{P}^o}^h \{\nu(L_1 + L_2 - D)\},$$

where the map to $\mathfrak{P}^o$ takes the normalized determinant of the direct sum of the sources. Its truncation W is the corresponding part of B^G , by the family decomposition just proved.

Proposition 3.6 (The actual fixed obstruction theory). *The enhancement $\mathfrak{W}$ induces a perfect obstruction theory $E_W \rightarrow L_W$. For $k : W \hookrightarrow B$, there is a natural isomorphism of obstruction theories*

$$(Lk^*E_B)^{\text{fix}} \xrightarrow{\sim} E_W, \quad (3.8)$$

compatible with their maps to L_W . Its virtual tangent is

$$E_W^\vee \simeq \text{fib} \left\{ \bigoplus_{i=1}^2 R\text{Hom}(S'_i, Q_i) \longrightarrow R\Gamma(\mathcal{I}_o)[1] \otimes \mathcal{O}_W \right\}. \quad (3.9)$$

The moving virtual normal complex is

$$N_W^{\text{vir}} = \bigoplus_{i \neq j} R\text{Hom}(S'_i, Q_j), \quad (3.10)$$

where the (i, j) summand has the difference between the characters on L_j and L_i , or evaluated weight $w_j - w_i$.

Proof. Direct sum is an actual morphism of perfect-arrow moduli problems. The determinant of a direct sum is the tensor product of determinants, compatibly with normalization. Consequently direct sum induces a derived map $\mathfrak{k} : \mathfrak{W} \rightarrow \mathfrak{B}$ lifting k . This map is coherently equivariant. Over the classifying stack BG , tensor the source and target of each summand arrow with the same character line of weight $+1$ or -1 . Normalizing the determinant cancels each base character line against its restriction at o . These canonical cancellations respect tensor products and the chosen homotopy in the determinant fiber. Thus direct sum defines $BG \times \mathfrak{W} \rightarrow [\mathfrak{B}/G]$ over BG , with trivial G -action on $\mathfrak{W}$. Relative cotangents over BG give its equivariant cotangent morphism. Its cotangent morphism, restricted to truncations, gives

$$Lk^*E_B \longrightarrow E_W \longrightarrow L_W.$$

By functoriality of the truncation comparison, this composite equals $Lk^*E_B \rightarrow Lk^*L_B \rightarrow L_W$.

To identify this map, use the tangent formula for arrows. On W ,

$$R\text{Hom}(S', Q) = \bigoplus_{i,j} R\text{Hom}(S'_i, Q_j).$$

Direct sum maps the two input tangents to the diagonal blocks. Trace vanishes on off-diagonal blocks, and the based Picard tangent has weight zero. To display the comparison, abbreviate these complexes by $\mathcal{A}_0 = \bigoplus_i R\text{Hom}(S'_i, Q_i)$, $\mathcal{A} = \bigoplus_{i,j} R\text{Hom}(S'_i, Q_j)$, and $\mathcal{P} = R\Gamma(\mathcal{I}_o)[1] \otimes \mathcal{O}_W$. The derivative gives the morphism of fiber triangles

$$\begin{array}{ccccc} E_W^\vee & \longrightarrow & \mathcal{A}_0 & \longrightarrow & \mathcal{P} \\ \downarrow & & \downarrow & & \parallel \\ Lk^*E_B^\vee & \longrightarrow & \mathcal{A} & \longrightarrow & \mathcal{P}, \end{array}$$

whose middle map is diagonal inclusion. After taking weight zero, the middle and right maps are identities. The left map is therefore an isomorphism of perfect complexes in families. The remaining summands are exactly (3.10).

In particular (3.9) has Tor-amplitude $[0, 1]$, as a direct summand of $Lk^*E_B^\vee$. Lemma 3.2 makes $E_W \rightarrow L_W$ a perfect obstruction theory. Dualizing the fiber-triangle comparison gives (3.8). The compatibility already established shows that this is the fixed obstruction morphism used in [GP99, Proposition 1], and not just an identity of virtual tangent classes. $\square$

The hypotheses of virtual localization are now in place: B is projective, Lemma 3.5 supplies an equivariant embedding and global bundle presentations, and Proposition 3.6 identifies the fixed obstruction morphism. Hence [GP99, Section 3] gives

$$[B]^{\text{vir}} = \sum_W k_* \frac{[W]^{\text{vir}}}{e_G(N_W^{\text{vir}})} \quad \text{in } \text{CH}_*^G(B) \otimes_{\mathbb{Q}[t]} \mathbb{Q}(t). \quad (3.11)$$

Here $[W]^{\text{vir}}$ is the ordinary virtual class just constructed, given its trivial equivariant lift. To see this explicitly, G acts trivially on W , and every equivariant coherent sheaf there decomposes into its weights. A weight-zero perfect theory is therefore the ordinary theory with trivial weights. On the mixed quotients defining equivariant Chow groups, its intrinsic cone and zero-section Gysin construction are the ordinary ones pulled back from W . This identifies its equivariant virtual class with that lift. Every weight in the normal complex is nonzero, so the Euler denominator is invertible after localization.

4 Fixed virtual classes and the divisor cosection

We compute the ordinary fixed virtual classes furnished by Proposition 3.6. Throughout this section, a symbol $\text{Div}^P(X)$ denotes the scheme of effective divisors whose line bundle lies in the specified component of $\text{Pic}(X)$; when that component is a point, this is the complete linear system $|P|$. The universal divisor is denoted by $\mathcal{C}$, and π always denotes projection along the surface X . All derived Hom complexes below include this derived projection. We retain the two settings of Section 3; in particular, $\chi(\mathcal{O}_X) = 1$ in both, and we put

$$m_P = \frac{P(P-K)}{2} = \chi(P) - 1.$$

The calculation has two different forms. When $p_g = q = 0$, each Picard component is a point and the based Picard tangent vanishes. The divisor factors are then complete linear systems, and the fixed loci are smooth; we compute their virtual classes from their obstruction bundles. On the auxiliary surface the divisor lines can vary in a Picard curve. We retain this variation in the obstruction theory, localize one divisor class to the empty or canonical divisor, and only then restrict the other factor and the insertions to the resulting fiber. The rank-one family decomposition and point-ideal comparison below provide the common input to both calculations.

4.1 Rank-one quotients in families

The divisor-and-point description and its tautological obstruction factors appear for a trivial rank-one target on a simply connected surface in [OP21, Section 5.2]. We record the factorization in arbitrary families and the obstruction-theory comparisons needed here.

We write $X^{[r]}$ for the Hilbert scheme of length- r zero-dimensional subschemes of X .

Lemma 4.1 (Divisor and point factorization). *Let R be a noetherian parameter scheme and let $L_R \rightarrow Q$ be a flat family of rank-zero quotients of a line bundle on $X \times R$. Its kernel has a unique factorization*

$$S' = L_R(-\mathcal{C})\mathcal{I}_Z,$$

where $\mathcal{C}$ is a relative effective Cartier divisor and $Z \subset X \times R$ is finite and flat over R . Both factors commute with arbitrary base change. Consequently, the rank-one Quot functor with specified divisor component and point length r is represented by

$$U = \text{Div}^P(X) \times X^{[r]}, \quad S' = L(-\mathcal{C})\mathcal{I}_Z. \quad (4.1)$$

The assertion includes nonreduced parameter schemes.

Proof. Untwist by L_R , and write $I \subset \mathcal{O}_{X \times R}$ for the kernel. It is flat over R , because the source and quotient are flat. Every geometric fiber I_u is torsion-free of rank one on a smooth surface and has projective dimension at most one. Locally choose a surjection from a vector bundle G onto I . Its kernel E is flat over R and has locally free geometric fibers, so is locally free by the local criterion of flatness. Thus

$$0 \longrightarrow E \longrightarrow G \longrightarrow I \longrightarrow 0$$

is a resolution by vector bundles, and remains a resolution after any base change.

Set $M = \det G \otimes (\det E)^{-1} = \det I$. If E has rank e , wedging a section of G with the image of $\bigwedge^e E$ gives the maximal-minor map $G \rightarrow M$; it kills E and induces $I \rightarrow M$. On the locus where I is invertible it is the canonical determinant identification. These local maps glue. To justify this last assertion over a nonreduced base, the fiber criterion for local freeness shows that the closed noninvertible locus has finite fibers over R ; properness makes this locus finite over R . Any closed support with finite fibers in the smooth relative surface has grade at least two locally: choose two elements of its ideal which form a regular sequence on the surface fiber and lift them; successive applications of the local flatness criterion show that they remain a regular sequence in the total space and that the successive quotients are base-flat. Powers of these elements annihilate any coherent sheaf with that support; the regular-sequence calculation therefore makes its sheaf $\mathcal{H}om$ and $\mathcal{E}xt^1$ into a line bundle vanish. In particular a section of a line bundle supported there is zero. Hence two local minor maps which agree on the invertible locus agree everywhere.

On each geometric fiber the resulting map $I_u \rightarrow M_u$ is the usual inclusion of a rank-one torsion-free sheaf into its determinant line: it is injective, and its cokernel has finite support. Fiberwise injectivity between base-flat modules now implies that $I \rightarrow M$ is injective with base-flat cokernel. After twisting by M^{-1} it is an ideal $J \subset \mathcal{O}_{X \times R}$, and $\mathcal{O}/J$ is base-flat with finite fibers. Properness of X makes its support finite over R . It therefore defines a finite flat subscheme Z , with $I = M\mathcal{I}_Z$.

The original inclusion $I \hookrightarrow \mathcal{O}_{X \times R}$ extends uniquely to a map $M \rightarrow \mathcal{O}_{X \times R}$. Indeed the grade-two observation, applied to the support of $M/I = M\mathcal{O}_Z$, gives

$$\mathcal{H}om(M\mathcal{O}_Z, \mathcal{O}_{X \times R}) = \mathcal{E}xt^1(M\mathcal{O}_Z, \mathcal{O}_{X \times R}) = 0.$$

Applying $\mathcal{H}om(-, \mathcal{O}_{X \times R})$ to $0 \rightarrow I \rightarrow M \rightarrow M\mathcal{O}_Z \rightarrow 0$ proves the extension and uniqueness. Its restriction to every surface fiber is nonzero, hence injective. The same flatness criterion makes it an injection with base-flat cokernel. It is therefore a relative effective Cartier divisor inclusion, $M = \mathcal{O}(-\mathcal{C})$.

The determinant resolutions, minor maps, and short exact sequences all survive base change, the last because their quotients are base-flat. The regular sequences used for uniqueness also survive base change, because their successive quotients are base-flat. Thus uniqueness identifies the resulting factorization after every base change, including a nonnoetherian one. Conversely, multiplying a relative Cartier ideal by a finite flat point ideal gives a flat quotient: its quotient is an extension of $\mathcal{O}_{\mathcal{C}}$ by $\mathcal{O}_Z(-\mathcal{C})$, both base-flat. This construction inverts the preceding one. More explicitly, any other line extending I with relatively finite cokernel is uniquely identified with M : the same grade-two extension argument extends the identity on I in both directions, and uniqueness makes the composites identities. This proves the functorial factorization, and hence the asserted isomorphism of parameter schemes. $\square$

Apply the Lemma to the two weight summands of a fixed quotient. Write $D_1 = T$, $D_2 = H$, and $T + H = D$ in the Picard scheme. If their point lengths are n_1, n_2 , comparison of the second

Chern characters gives

$$n_1 + n_2 + L_1 T + L_2 H - \frac{T^2 + H^2}{2} = n + L_1 D - \frac{D^2}{2}.$$

Since $A = L_1 - L_2$, this is

$$n_1 + n_2 = n - (D - A)H + H^2. \quad (4.2)$$

These labels form a finite set on the fixed locus under consideration: the Picard components, Hilbert polynomials, and point lengths are locally constant there, and the fixed locus is of finite type.

4.2 The point-ideal complexes

The fixed locus has been expressed in divisor and point factors. We next identify the point contribution to its obstruction theory; this will separate the Hilbert-scheme tangent from the excess bundle.

For a line bundle P , write $P^{[r]} = \pi_* \mathcal{O}_Z(P)$ for its tautological bundle of rank r on $X^{[r]}$. For the universal ideal $\mathcal{I}$, the trace-free endomorphism complex means the fiber of $\text{tr} : R\text{Hom}(\mathcal{I}, \mathcal{I}) \rightarrow R\Gamma(\mathcal{O}_X)$.

Lemma 4.2 (Trace splitting and the point obstruction). *On $X^{[r]}$, with universal ideal $\mathcal{I}$, there is a quasi-isomorphism*

$$R\text{Hom}(\mathcal{I}, \mathcal{I}) \simeq R\Gamma(\mathcal{O}_X) \oplus T_{X^{[r]}}[-1]. \quad (4.3)$$

The two maps to the right-hand side are the trace and the inverse of the trace-free Kodaira–Spencer isomorphism. On $\text{Div}^P(X) \times X^{[r]}$, define

$$F = (\pi_* \mathcal{O}_Z(K - \mathcal{C}))^\vee. \quad (4.4)$$

It is a vector bundle of rank r , and relative Serre duality gives

$$R\text{Hom}(\mathcal{O}_Z, \mathcal{O}(\mathcal{C})) = F[-2]. \quad (4.5)$$

Proof. The scalar map $R\Gamma(\mathcal{O}_X) \rightarrow R\text{Hom}(\mathcal{I}, \mathcal{I})$ is split by trace, since $\mathcal{I}$ has rank one. At a geometric point Z , extending across its codimension-two support gives

$$\text{Hom}(I_Z, I_Z) = \mathbb{C}, \quad \text{Hom}(I_Z, I_Z(K)) = H^0(K).$$

For example, an endomorphism extends to the determinant line and is there multiplication by a global section; conversely multiplication preserves the ideal. By Serre duality the trace-free $\text{Ext}^2(I_Z, I_Z)$ vanishes, as does the trace-free Ext^0 . Riemann–Roch gives $\chi(I_Z, I_Z) = \chi(\mathcal{O}_X) - 2r$; hence the trace-free Ext^1 has dimension $2r$.

The boundary map of the point-ideal sequence embeds

$$T_Z X^{[r]} = \text{Hom}(I_Z, \mathcal{O}_Z) \longrightarrow \text{Ext}^1(I_Z, I_Z).$$

Indeed every map $I_Z \rightarrow \mathcal{O}_X$ is a scalar multiple of inclusion, so its composite with $\mathcal{O}_X \rightarrow \mathcal{O}_Z$ is zero. This boundary is the Kodaira–Spencer map of the universal ideal. Its trace vanishes, because the universal point ideal has constant determinant. Fogarty’s smoothness Theorem [Fog68, Theorem 2.4] gives $\dim X^{[r]} = 2r$, so this is an isomorphism onto the trace-free part. The family complexes are perfect and commute with derived base change. A map between them is a quasi-isomorphism if it is so on geometric fibers, proving (4.3).

For the other assertion, Z is finite flat of length r , so $\pi_* \mathcal{O}_Z(K - \mathcal{C})$ is locally free of that rank, with arbitrary base change and no higher direct images. Relative Serre duality for the smooth proper surface projection identifies the dual of $R\text{Hom}(\mathcal{O}_Z, \mathcal{O}(\mathcal{C}))$ with $\pi_* \mathcal{O}_Z(K - \mathcal{C})[2]$. Dualizing gives precisely (4.5), including its shift and the indicated dual bundle. $\square$

For later calculations it follows that the virtual tangent class of a rank-one arrow is

$$[T_U^{\text{vir}}] = [R\pi_* \mathcal{O}(\mathcal{C})] - [F] - [R\Gamma(\mathcal{O}_X)] + [T_{X^{[r]}}]. \quad (4.6)$$

In fact it is the cone of $R\text{Hom}(\mathcal{I}, \mathcal{I}) \rightarrow R\text{Hom}(\mathcal{I}, \mathcal{O}(\mathcal{C}))$, and the ideal sequence and (4.5) give the formula.

Lemma 4.3 (Euler class for a smooth morphism). *Suppose $g : Y \rightarrow Z$ is smooth and carries a relative perfect obstruction theory $E_g \rightarrow L_{Y/Z}$ of amplitude $[-1, 0]$. Then $\text{Ob}_g = h^1(E_g^\vee)$ is a vector bundle,*

$$[\text{Ob}_g] = [T_{Y/Z}] - [E_g^\vee] \quad \text{in } K^0(Y),$$

and its virtual pullback is

$$g^! \gamma = e(\text{Ob}_g) \cap g^* \gamma. \quad (4.7)$$

The formula holds over arbitrary base cycles. For Z a point, it says $[Y]^{\text{vir}} = e(\text{Ob}_Y) \cap [Y]$.

Proof. Here $L_{Y/Z} = \Omega_{Y/Z}$ is a vector bundle in degree zero. Locally split the surjection from a two-term presentation of E_g onto this bundle. The remaining acyclic degree-zero terms can be cancelled; one obtains

$$E_g \simeq \Omega_{Y/Z} \oplus \text{Ob}_g^\vee[1],$$

with comparison map the projection. This shows local freeness and the stated K -class, without requiring a global splitting. The relative intrinsic normal cone of a smooth map is $[0/T_{Y/Z}]$. The dual comparison morphism gives its global inclusion into $h^1/h^0(E_g^\vee)$, with quotient bundle Ob_g . Locally this is the inclusion into $[\text{Ob}_g/T_{Y/Z}]$ supplied by the splitting above; globally the subbundle excess formula gives its zero-section Gysin image as the Euler class of Ob_g . Applying this excess formula over each base cycle proves (4.7). $\square$

4.3 The fixed class when $p_g = q = 0$

Proposition 4.4. *Assume $p_g = q = 0$. A nonempty fixed part with lines T, H and lengths satisfying (4.2) is the smooth product*

$$W = |T| \times |H| \times X^{[n_1]} \times X^{[n_2]}.$$

Let $x_i = c_1(\mathcal{O}_{|D_i|}(1))$, with $D_1 = T, D_2 = H$. The fixed virtual class is

$$[W]^{\text{vir}} = \prod_{i=1}^2 (x_i^{h^1(D_i)} e(F_i)) \cap [W], \quad F_i = ((K - D_i)^{[n_i]})^\vee \otimes \mathcal{O}(1_i). \quad (4.8)$$

A part with either $m_T < 0$ or $m_H < 0$ contributes zero. For the other parts, after including $e(F_1) e(F_2)$, pushforward over the linear systems is coefficient extraction at $[x_1^{m_T} x_2^{m_H}]$.

Proof. The Picard scheme is discrete and reduced, and $R\Gamma(\mathcal{I}_o) = 0$. Thus the determinant condition leaves just the indicated products, with the sum of their two rank-one obstruction theories. For an effective line P , multiplication by any nonzero section of P injects $H^0(K - P)$ into $H^0(K) = 0$, so $H^2(P) = 0$. On its linear system, $\mathcal{O}(\mathcal{C}) = P \boxtimes \mathcal{O}(1)$. The Euler sequence and (4.6) therefore give

$$[T_{|P| \times X^{[r]}}] - [T_U^{\text{vir}}] = [H^1(P) \otimes \mathcal{O}(1)] + [F].$$

Apply Lemma 4.3. Equality of these vector bundle classes gives equality of their Chern classes, and hence (4.8); it does not assert a global splitting of the obstruction bundle.

Finally,

$$\dim |P| - h^1(P) = h^0(P) - 1 - h^1(P) = m_P.$$

If $m_P < 0$, the factor $x^{h^1(P)}$ is already zero in $\mathrm{CH}^*(|P|)$. Otherwise integration against that factor replaces extraction at degree $\dim |P|$ by extraction at degree m_P . The localized moving factors are expanded with nonzero equivariant constant terms, so introduce no negative hyperplane powers. Thus this argument applies also to the localized insertions used below. $\square$

4.4 A compatible relative theory in the auxiliary setting

We now assume $p_g = q = 1$, that K is ample, and that $D = 3K$. We impose the determinant constraint in one rank-one factor and apply the divisor cosection to the other. On each fixed part choose an index a such that $KD_a > K^2$, which is possible since $KD_1 + KD_2 = 3K^2$. Call it the high index, and call the other index b the low index. If both satisfy the inequality, choose either one. The ensuing vanishing argument applies to the resulting low index as well.

Lemma 4.5 (Relative determinant pullback). *For the projection $f : W \rightarrow U_b$ to the unrestricted low rank-one Quot, there is a relative perfect obstruction theory whose virtual tangent is*

$$T_f^{\mathrm{vir}} = \mathrm{fib}\{R\mathrm{Hom}(S'_a, Q_a) \longrightarrow R\Gamma(\mathcal{I}_o)[1]\}. \quad (4.9)$$

It is compatible with the absolute theories of W and U_b , and

$$[W]^{\mathrm{vir}} = f^![U_b]^{\mathrm{vir}}. \quad (4.10)$$

Proof. Use the actual projection of enhancements $\mathfrak{W} \rightarrow \mathfrak{U}_b$ from Proposition 3.6. The tangent of its determinant fiber in the high direction is (4.9). The map there is the high arrow's boundary followed by trace and based normalization. Lemma 3.3, with rank one and $KD_a > K^2$, makes its degree-one map onto $H^2(\mathcal{O}_X)$ surjective, so the fiber has amplitude $[0, 1]$. Explicitly, an annihilating canonical form would annihilate Q_a ; at each generic curve point a quotient of one generator annihilated by that form has length at most its vanishing order. Thus $KD_a \leq K^2$, a contradiction.

Restrict the derived cotangent transitivity triangle to W . It has a natural morphism to the classical transitivity triangle, giving

$$\begin{array}{ccccc} Lf^*E_{U_b} & \longrightarrow & E_W & \longrightarrow & E_f \\ \downarrow & & \downarrow & & \downarrow \\ Lf^*L_{U_b} & \longrightarrow & L_W & \longrightarrow & L_{W/U_b}. \end{array}$$

Each row continues by a shift of its first term. Here the absolute comparisons have cones in degrees at most -2 , by Lemma 3.2. If these cones are C_{U_b} and C_W , and the relative comparison cone is C_f , the diagram gives a triangle

$$Lf^*C_{U_b} \longrightarrow C_W \longrightarrow C_f \longrightarrow Lf^*C_{U_b}[1].$$

Derived pullback preserves the upper cohomological bound, and the shift $[1]$ lowers that bound. Hence C_f is also concentrated in degrees at most -2 . Combined with the amplitude calculation, this proves that E_f is a relative perfect obstruction theory.

We have thus produced the compatible triple, including its morphisms to the cotangent triangle, required in [Man12, Definition 4.5]. The morphisms are between quasiprojective schemes, so are of Deligne–Mumford type and have the required stratifications by global quotients. Virtual-pullback functoriality [Man12, Theorem 4.8 and Corollary 4.9] gives (4.10). $\square$

4.5 The point obstruction as an excess bundle

Let $\mathfrak{D}^P$ denote the divisor enhancement: the open rank-one arrow stack with invertible source and divisor quotient. Its classical truncation is $\mathrm{Div}^P(X)$. For a divisor C , with source $M = L(-C)$, its tangent is

$$R\mathrm{Hom}(M, L/M) = R\pi_* \mathcal{O}_C(C).$$

It has amplitude $[0, 1]$, and its Euler characteristic is $\chi(P) - \chi(\mathcal{O}_X) = m_P$. Thus it induces a divisor virtual class of dimension m_P .

Lemma 4.6 (Compatible point excess). *On the product $U_b = \mathrm{Div}^{D_b}(X) \times X^{[n_b]}$, the rank-one Quot virtual class is*

$$[U_b]^{\mathrm{vir}} = e(F_b) \cap ([\mathrm{Div}^{D_b}(X)]^{\mathrm{vir}} \times [X^{[n_b]}]). \quad (4.11)$$

Proof. There is a morphism of derived moduli problems

$$g : \mathfrak{D}^{D_b} \times X^{[n_b]} \longrightarrow \mathfrak{U}_b, \quad (M_b \rightarrow L_b, \mathcal{I}_b) \longmapsto (M_b \mathcal{I}_b \rightarrow M_b \rightarrow L_b). \quad (4.12)$$

The ordinary smooth Hilbert scheme here carries its smooth enhancement, and its universal ideal is perfect. Tensoring and composing perfect families define the displayed map on derived bases as well. Lemma 4.1 identifies its truncation with the identity of U_b .

To compute its differential, first forget the arrows and compare their source moduli. On the common truncation the map is

$$R\Gamma(\mathcal{O}_X)[1] \oplus T_{X^{[n_b]}} \longrightarrow R\mathrm{Hom}(\mathcal{I}_b, \mathcal{I}_b)[1]. \quad (4.13)$$

The first summand acts by scalars and the second is the Kodaira–Spencer map of the ideal, so Lemma 4.2 makes this a quasi-isomorphism. Indeed, on each geometric fiber the scalar map identifies the trace summands, and the Hilbert map induces an isomorphism onto the trace-free Ext^1 . The source derivative is therefore triangular with isomorphic diagonal blocks. Its perfect cone has zero geometric fibers, so vanishes; no choice of a global splitting is needed here. The scalar assertion follows directly by differentiating tensoring with a line. Before taking sections it is $\mathcal{O}_X[1] \rightarrow R\mathrm{Hom}(\mathcal{I}_b, \mathcal{I}_b)[1]$, and corresponds to the identity endomorphism.

The relative arrow tangent map over these source moduli is

$$R\mathrm{Hom}(M_b, L_b) \longrightarrow R\mathrm{Hom}(M_b \mathcal{I}_b, L_b).$$

The point-ideal sequence and (4.5) identify its cone with

$$R\mathrm{Hom}(\mathcal{O}_{Z_b}, \mathcal{O}(\mathcal{C}_b))[1] = F_b[-1].$$

The comparison of the two source tangent triangles, using (4.13), therefore proves

$$T_{\mathrm{prod}}^{\mathrm{vir}} \longrightarrow T_{U_b}^{\mathrm{vir}} \longrightarrow F_b[-1] \longrightarrow T_{\mathrm{prod}}^{\mathrm{vir}}[1]. \quad (4.14)$$

Crucially, dualizing this is a morphism of perfect obstruction theories $E_{U_b} \rightarrow E_{\mathrm{prod}}$, compatible with their maps to the same L_{U_b} , because it came from (4.12). Thus the induced map of obstruction bundle stacks is an inclusion

$$h^1/h^0(T_{\mathrm{prod}}^{\mathrm{vir}}) \hookrightarrow h^1/h^0(T_{U_b}^{\mathrm{vir}})$$

with quotient vector bundle F_b . For precision, on an affine open the extension in (4.14) splits: its class lies in $\mathrm{Hom}(F_b, T_{\mathrm{prod}}^{\mathrm{vir}}[2]) = 0$, since F_b is locally free and the tangent has amplitude $[0, 1]$. This describes the inclusion locally as a vector bundle substack; the quotient is globally the displayed F_b .

Compatibility of the obstruction morphisms implies that the intrinsic normal cone embedded using the Quot theory is the image of its embedding using the product theory. The subbundle excess formula for zero-section Gysin maps therefore multiplies the latter virtual class by $e(F_b)$. One can see the formula on any base cycle by pulling to the larger bundle stack: the smaller stack is the zero locus of the quotient-bundle coordinate, so its inclusion multiplies the pullback class by the top Chern class of that quotient. This is the zero-section excess formula of [Ful98, §6.3]; its vector-bundle form is also [KL13, Lemma 2.5]. We apply it to the bundle-stack presentations of the intrinsic cone in [BF97]. The product theory has class $[\mathrm{Div}^{D_b}(X)]^{\mathrm{vir}} \times [X^{[n_b]}]$. This proves (4.11) on the possibly singular divisor scheme. $\square$

4.6 Localization on the divisor scheme

The excess comparison separates the low point factor from its divisor virtual class. We now determine the support of that divisor class and the sign of its canonical contribution, using the divisor cosection studied by Chang–Kiem [CK13, Section 3.2].

Lemma 4.7 (Support of the divisor class). *In the auxiliary setting a divisor component with $m_P < 0$ has zero virtual class. If $m_P \geq 0$, its virtual class is supported, by cosection localization, on the empty divisor or on the unique canonical divisor z . The empty-divisor class is the reduced point with coefficient one. Only the Picard component containing K can give the canonical contribution.*

Proof. The first assertion is the vanishing of a Chow group in negative dimension. For the second, the universal divisor sequence yields a cosection

$$\sigma : \mathrm{Ob}_{\mathrm{Div}^P} = R^1\pi_*\mathcal{O}_C(C) \longrightarrow H^2(\mathcal{O}_X) \otimes \mathcal{O} \simeq \mathcal{O}. \quad (4.15)$$

At a divisor C of line class P the long exact cohomology sequence ends in

$$H^1(\mathcal{O}_C(C)) \xrightarrow{\sigma_C} H^2(\mathcal{O}_X) \longrightarrow H^2(P) \longrightarrow 0.$$

The obstruction fibers agree with these groups because the perfect tangent has amplitude $[0, 1]$. Thus $\mathrm{coker}(\sigma_C) = H^2(P) = H^0(K - P)^\vee$. The cosection is not surjective exactly where $K - P$ is effective. Kiem–Li localization [KL13, Theorem 1.1] expresses the ordinary virtual class as the pushforward of a class on this degeneracy locus.

Suppose a degeneracy point exists and put $r = KP$. Both P and $K - P$ are effective. Since K is ample and $m_P \geq 0$, Hodge index gives

$$0 \leq r \leq K^2, \quad r \leq P^2 \leq \frac{r^2}{K^2}.$$

For $0 < r < K^2$ these inequalities contradict one another. If $r = 0$, ampleness forces the effective divisor C to be empty. If $r = K^2$, any effective divisor in $K - P$ has zero intersection with K , so is empty. In the latter case $P \cong K$ as line bundles, and $h^0(K) = 1$ gives exactly one divisor z .

For the empty divisor the tangent complex $R\pi_*\mathcal{O}_\emptyset$ is zero. Its parameter scheme is therefore a reduced point with the ordinary point class. A nontrivial numerically trivial line has no effective divisor, again by ampleness. This proves all the support assertions. $\square$

Lemma 4.8 (Canonical multiplicity and sign). *For the divisor component whose Picard component contains K , there is an integer $s > 0$ such that*

$$[\mathrm{Div}^K(X)]^{\mathrm{vir}} = -s[z]. \quad (4.16)$$

Here $\mathrm{Div}^K(X)$ allows variation of the line in its Picard component; z is the unique effective canonical divisor.

Proof. Let C_{Pic} be the Picard component containing K ; it is a smooth projective curve because $q = 1$. Choose an open neighborhood V of the point K . Upper semicontinuity, followed by shrinking, gives $h^0(P) \leq 1$ for $P \in V$. For $P \neq K$ in this component, $K - P$ is a nontrivial numerically trivial line, so $h^2(P) = 0$. Riemann–Roch gives $\chi(P) = 1$, and consequently $h^0(P) = 1 + h^1(P) \geq 1$. At K , $h^0(K) = 1$ by assumption. Thus $h^0(P)$ is constantly one on V .

We verify the resulting scheme structure, including infinitesimal families. Represent the derived direct image of the based universal line over V , locally, by vector bundles

$$E^0 \xrightarrow{d^0} E^1 \xrightarrow{d^1} E^2.$$

There are no negative terms because this is the direct image of a line bundle, and relative dimension two bounds the positive terms. The kernel of the fiber of d^0 has constant dimension one. Since V is reduced, all minors above the constant rank vanish scheme-theoretically. After inverting a maximal nonzero minor, elementary row and column operations turn d^0 into an identity block and a zero block. Its kernel is therefore a line subbundle and commutes with arbitrary, including nonreduced, base change. It is the bundle of sections of the universal line. The divisor functor over V is its projective bundle of nonzero sections up to scalar, and hence is V itself. Thus the divisor scheme is a smooth curve in a neighborhood of z .

The divisor theory has virtual dimension $m_K = 0$. Lemma 4.3 makes its obstruction sheaf an obstruction line $\mathcal{E}$ on this neighborhood. At z the map $H^2(\mathcal{O}_X) \rightarrow H^2(K)$ in the divisor sequence is an isomorphism: its dual maps 1 to the nonzero canonical section. Hence the cosection $\mathcal{E} \rightarrow \mathcal{O}_V$ vanishes at z . By Lemma 4.7 it is nonzero at every other point nearby. On the smooth curve its germ is multiplication by a unit times a uniformizer to a finite positive power s . It follows that the line with its cosection is isomorphic to

$$(\mathcal{O}_V(-sz), \mathcal{O}_V(-sz) \hookrightarrow \mathcal{O}_V).$$

Shrink to an affine neighborhood of z . As in Lemma 4.3, the perfect obstruction theory there splits, with its comparison map, as $\Omega_V \oplus \mathcal{E}^\vee[1] \rightarrow \Omega_V$. The localized intrinsic-cone calculation is thus the localized Euler class of the zero section in $\mathcal{E} = \mathcal{O}_V(-sz)$. For clarity, in the regularizing formula [KL13, Definition 2.3 and equation (2.1)], take the identity modification of V , the quotient line $\mathcal{O}_V(D)$ with $D = -sz$, and its zero kernel. The lifted zero cone is V , so the formula gives

$$[D] \cap [V] = -s[z].$$

This determines the coefficient of the global localized class: localized Gysin maps commute with restriction to an open neighborhood of the degeneracy locus, by restricting the regularizing modification, its cone cycles, and the ordinary Gysin maps. This excision is also stated in [CK13, Proposition 2.3]. The support lemma shows that z is the only degeneracy point on the whole divisor component. Pushing its localized class forward, by [KL13, Theorem 1.1], proves (4.16). The argument used smoothness of the Picard curve; it imposed no smoothness or reducedness condition on the canonical divisor itself. $\square$

Chang and Kiem determine the canonical-divisor virtual coefficient more precisely as $(-1)^{\chi(\mathcal{O}_X)}$ for minimal surfaces of general type with $p_g > 0$ [CK13, Theorem 1.3 and Corollary 3.4]. Their divisor theory has obstruction sheaf $R^1\pi_*\mathcal{O}_{\mathcal{C}}(\mathcal{C})$ [CK13, Theorem 3.1], and its cosection is (4.15) after identifying $H^2(\mathcal{O}_X)$ with $\mathbb{C}$ using a nonzero canonical form [CK13, Section 3.2]. On the smooth neighborhood V above, both theories therefore give the localized Euler class of the same obstruction line and cosection. Excision identifies their coefficients, so $s = 1$ here. The local argument above retains only the sign and nonvanishing needed here. This sign comes from the divisor cosection; the moving Euler classes have their own weight signs, computed separately in Section 5.

4.7 The high factor and restriction to its fibers

Proposition 4.9. *Assume $p_g = q = 1$, K ample, and $D = 3K$. Every fixed contribution vanishes except those obtained from a low divisor equal to the empty divisor or to the canonical divisor. In either surviving case the contribution is computed on the actual fiber*

$$|D_a| \times X^{[n_a]} \times X^{[n_b]},$$

with the factors $e(F_a)e(F_b)$. The low hyperplane is $x_b = 0$ and $m_{D_b} = 0$; pushforward over $|D_a|$ is extraction at $x_a^{m_{D_a}}$. The overall multiplier is 1 for the empty divisor and $-s$, with the same $s > 0$ as in (4.16), for the canonical divisor. The statement applies to operational insertions in universal perfect complexes, also after adjoining marked surface factors.

Proof. By Lemmas 4.5 and 4.6, the fixed class is obtained by applying $f^!$ to

$$e(F_b) \cap ([\mathrm{Div}^{D_b}(X)]^{\mathrm{vir}} \times [X^{[n_b]}]).$$

The support and canonical-sign Lemmas make this input zero unless its divisor class is the point class of the empty divisor or $-s[z]$.

Over each of the corresponding low Picard components, the high line determined by the determinant has $D_a - K$ numerically $2K$ or K , respectively. It is therefore ample. Kodaira vanishing gives $H^j(D_a) = 0$ for $j > 0$, uniformly in that component. Use the based universal low line to obtain the based high line as $D - D_b$. Its pushforward is a vector bundle of rank $\chi(D_a) = 1 + m_{D_a}$, with arbitrary base change. Lemma 4.1 consequently expresses f here as its relative projective bundle of high divisors, followed by the product with the smooth Hilbert scheme $X^{[n_a]}$. In particular f is smooth, even if the low divisor scheme away from the degeneracy point is singular.

Formula (4.9), together with (4.6), gives

$$\begin{aligned} [T_f^{\mathrm{vir}}] &= [R\pi_* \mathcal{O}(\mathcal{C}_a)] - [F_a] + [T_{X^{[n_a]}}] \\ &\quad - [R\Gamma(\mathcal{O}_X)] - [R\Gamma(\mathcal{I}_o)[1]]. \end{aligned}$$

The evaluation sequence at o implies

$$[R\Gamma(\mathcal{O}_X)] + [R\Gamma(\mathcal{I}_o)[1]] = 1.$$

Thus the two determinant and scalar terms together subtract exactly one scalar. Higher cohomology vanishes for the high line, and the relative projective Euler sequence now identifies

$$[T_f^{\mathrm{vir}}] = [T_f] - [F_a].$$

Lemma 4.3 proves, on all base cycles,

$$f^! \gamma = e(F_a) \cap f^* \gamma. \tag{4.17}$$

Here the use of the K -class is legitimate: the compatible relative perfect obstruction theory has already been constructed, and smoothness identifies its virtual operation with the Euler class of a vector bundle. Its Chern classes are determined by the displayed K -class.

Apply (4.17) to the point-supported low input class. Compatibility of smooth pullback with closed pushforward restricts it to the fiber over that low divisor point, while retaining $X^{[n_b]}$. The projection formula similarly restricts every operational insertion to the fiber. The universal divisors there are $D_i \boxtimes \mathcal{O}(1_i)$, with the low factor a point; in particular $x_b = 0$. The high projective space has dimension $h^0(D_a) - 1 = m_{D_a}$, proving the extraction rule. The universal point ideals, quotient complexes, and their derived Hom complexes restrict by derived base change for the smooth proper surface projection. All of these operations commute with adjoining the marked surface factors. This proves the asserted fiber formula with the multipliers already calculated. $\square$

5 A Chow relation and its universal diagonal coefficient

Our goal is the relation $0 = c[\Delta_X] + \Gamma$ of (1.1), with Γ a sum of external products. We first obtain a vanishing by equivariant degree, then decompose each fixed contribution into a diagonal term and external products, retaining both surface factors.

Throughout this Section Chow groups have rational coefficients. We use the equivariant perfect obstruction theory of Proposition 3.4, its fixed theory in Proposition 3.6, and the fixed virtual classes computed in Propositions 4.4 and 4.9. Both settings have $\chi(\mathcal{O}_X) = 1$. The torus parameter is initially the indeterminate t , of equivariant codimension one; the weights on L_1, L_2 are $t, -t$. Only after localization will we set $t = 1/2$, obtaining $w_1 = 1/2, w_2 = -1/2$.

5.1 The vanishing supplied by equivariant degree

Fix $p \geq 0$ such that $v > p + 2$, where $v = 2n + (A - K)D$ is the virtual dimension in (3.7). Let Q also denote the perfect universal quotient on $B \times X$. For the point $o \in X$ used to normalize the determinant, define

$$h = c_1^{\mathbb{G}_m}(\mathbf{L}^*_{i_o} Q) \quad \text{in } \mathrm{CH}_{\mathbb{G}_m}^1(B), \quad i_o : B \longrightarrow B \times X.$$

The derived restriction in this definition is essential to make it an operation on the perfect universal family, even where the quotient is not locally free. Let $q : B \times X \times X \rightarrow X \times X$ be projection, and let the subscripts (1), (2) mean pullback from the two copies of $B \times X$. Consider the nonlocalized equivariant class

$$\Xi(t) = q_* \left(h^p \mathrm{ch}_2^{\mathbb{G}_m}(Q)_{(1)} \mathrm{ch}_2^{\mathbb{G}_m}(Q)_{(2)} \cap ([B]^{\mathrm{vir}} \times [X \times X]) \right). \quad (5.1)$$

All insertions are Chern-class operations of perfect complexes, and q is proper. The total equivariant codimension of $\Xi(t)$ is $p + 4 - v$. Since the torus acts trivially on $X \times X$,

$$\mathrm{CH}_{\mathbb{G}_m}^*(X \times X) = \mathrm{CH}^*(X \times X)[t].$$

Consequently the coefficient of ordinary codimension two in the localized class would have to be accompanied by

$$t^{p+2-v}. \quad (5.2)$$

Its exponent is negative, whereas (5.1) has no negative powers of t . This component is therefore zero. Virtual localization [GP99] expresses the same class as a sum of fixed pushforwards divided by the moving Euler classes. Thus their ordinary codimension-two components sum to zero, and this remains true after $t = 1/2$. Each individual fixed pushforward has the same total equivariant codimension $p + 4 - v$: its ordinary codimension-two part is therefore t^{p+2-v} times a class independent of t . Consequently changing t to $-t$ multiplies each such contribution by $(-1)^{p+2-v}$. This termwise statement will allow us to compare the two orders of a fixed decomposition in Section 7.

5.2 A fixed contribution, including its signs

We first write an individual fiber contribution without the multiplier from the divisor virtual class. Fix ordered lines $D_1 = T, D_2 = H$ with $T + H = D$, and point lengths $n_i \geq 0$ satisfying (4.2). Put

$$M = X^{[n_1]} \times X^{[n_2]}, \quad m_i = m_{D_i} = \frac{D_i(D_i - K)}{2}.$$

The variables x_i are the hyperplane classes on the divisor linear systems; they can also be regarded as formal variables in the formula below. Write $\mathcal{I}_i$ for the universal ideals pulled to $M \times X$, and Z_i for their universal subschemes. For a line P let $P^{[n_i]}$ be its rank- n_i tautological bundle.

For a vector bundle E of rank r , our Euler convention is

$$e_u(E) = \sum_{j=0}^r c_j(E) u^{r-j}.$$

For a virtual bundle it is extended multiplicatively whenever u has nonzero constant equivariant part. Equivalently, $e_u(E) = u^{\text{rk } E} c_{1/u}(E)$, expanded in positive Chow degree. The fixed point-obstruction factors are

$$\mathcal{E}(x) = \prod_{i=1}^2 e_{x_i} \left(((K - D_i)^{[n_i]})^\vee \right). \quad (5.3)$$

Indeed $F_i = ((K - D_i)^{[n_i]})^\vee \otimes \mathcal{O}(1_i)$ on the indicated product or fiber, so (5.3) is $\prod_i e(F_i)$.

Define perfect K -classes on M by

$$\begin{aligned} P_{ij} &= L_j - L_i + D_i, \\ B_{ij} &= L_j - L_i + D_i - D_j, \\ E_{ij}(P) &= [R\Gamma(P)] - [R\text{Hom}(\mathcal{I}_i, \mathcal{I}_j(P))], \quad i \neq j. \end{aligned}$$

Here and below relative $R\text{Hom}$ includes derived pushforward along X . The class $E_{ij}(P)$ is the standard Ext K -class of Carlsson–Okounkov [CO12, Section 1.2, equation (2)]. The moving ratio below is derived here; their vertex-operator evaluation theorem is not used. The inverse Euler class of the moving virtual normal bundle is

$$\mathcal{R}(x) = \prod_{i \neq j} \frac{(w_j - w_i + x_i - x_j)^{\chi(B_{ij})} e_{w_j - w_i + x_i} \left(((K - P_{ij})^{[n_i]})^\vee \right)}{(w_j - w_i + x_i)^{\chi(P_{ij})} e_{w_j - w_i + x_i - x_j}(E_{ij}(B_{ij}))}. \quad (5.4)$$

We derive the signs explicitly. On the fiber under consideration,

$$S'_i = L_i(-D_i)\mathcal{I}_i \otimes \mathcal{O}(-1_i).$$

The ideal sequence and relative Serre duality give

$$[R\text{Hom}(\mathcal{I}_i, P)] = [R\Gamma(P)] - \left[((K - P)^{[n_i]})^\vee \right].$$

The duality shift is $[-2]$, so it contributes with a positive sign before subtraction by the ideal sequence. Put $u = w_j - w_i$. The moving class for the ordered pair (i, j) , namely $R\text{Hom}(S'_i, L_j) - R\text{Hom}(S'_i, S'_j)$, is therefore

$$\begin{aligned} & [R\Gamma(P_{ij})]_{u+x_i} - \left[((K - P_{ij})^{[n_i]})^\vee \right]_{u+x_i} \\ & - [R\Gamma(B_{ij})]_{u+x_i-x_j} + [E_{ij}(B_{ij})]_{u+x_i-x_j}. \end{aligned}$$

The subscript denotes the specified weight and line twist. A constant complex of vector spaces contributes the indicated weight raised to its Euler characteristic. Taking the inverse Euler class of this four-term expression proves (5.4); in particular E_{ij} occurs in the denominator. No individual cohomology dimension of P_{ij} or B_{ij} remains in the result.

For the Chern-character calculation, temporarily read w_1, w_2 as the degree-one equivariant classes $t, -t$: take the equivariant codimension-two component before specializing $t = 1/2$. The restriction of the quotient Chern character to a marked surface factor then follows from

$$\begin{aligned}\mathrm{ch}(Q_i) &= e^{L_i + w_i} \left(1 - e^{-D_i - x_i} \mathrm{ch}(\mathcal{I}_i) \right), \\ \mathrm{ch}(\mathcal{I}_i) &= 1 - [Z_i] + \text{terms of codimension at least three.}\end{aligned}$$

Its first Chern class is $D_i + x_i$. The determinant of the universal point ideal is trivial, so derived restriction to o adds no incidence term to this first Chern class. Line classes from X restrict to zero in $\mathrm{CH}^1(B)$, and the quotient has rank zero. Thus h restricts to $x_1 + x_2$. In particular, fixing the determinant in the Picard scheme does not make h zero. On this fixed product the determinant line of the universal quotient is $D \boxtimes \mathcal{O}(1_1 + 1_2)$, up to a constant line. Its based normalization removes the factor from the parameter space, whereas restriction of the unnormalized line to o retains that factor. This is exactly the distinction in Equation (3.5). The second Chern character on $M \times X$ is

$$J = \sum_{i=1}^2 \left([Z_i] + (L_i + w_i)(D_i + x_i) - \frac{(D_i + x_i)^2}{2} \right). \quad (5.5)$$

Let $\pi : M \times X \times X \rightarrow X \times X$ be projection. The ordinary codimension-two fiber contribution is the following class, where capping with $[M \times X \times X]$ in the pushforward is implicit:

$$\begin{aligned}C_*(T, H; n_1, n_2) &= \left[[x_1^{m_T} x_2^{m_H}] \right. \\ &\quad \left. \pi_*(\mathcal{E}(x)\mathcal{R}(x)(x_1 + x_2)^p J_{(1)} J_{(2)}) \right]_2.\end{aligned} \quad (5.6)$$

The outer subscript means ordinary codimension two on $X \times X$. All denominators are expanded at $x_1 = x_2 = 0$ and in positive Chow degree. Their constant weights are ± 1 , so these expansions are well defined. Only finitely many terms can contribute to the specified x -coefficient and Chow degree. A negative value of either m_i means that the contribution is zero.

For $p_g = q = 0$, Proposition 4.4 inserts $x_i^{h^1(D_i)} e(F_i)$ on $|D_i|$. Since $h^2(D_i) = 0$ for effective D_i and $\dim |D_i| = m_i + h^1(D_i)$, projective integration is precisely the coefficient extraction in (5.6). For the auxiliary setting, Proposition 4.9 first restricts to a fiber over the empty or canonical low divisor. The low variable is $x_b = 0$, its m_b is zero, and the high projective factor has dimension m_a . The same formula applies, with multiplier 1 or $-s$, respectively. This restriction precedes every calculation with the insertions; there is no remaining integration over a Picard parameter in (5.6).

5.3 Universality with both output factors retained

Let $R_X \subset \mathrm{CH}^*(X)$ denote the graded subring generated by $K, c_2(T_X), L_1, L_2, T, H$ and the identity. We shall show that (5.6) lies in

$$\mathbb{Q} \cdot [\Delta_X] + \sum_{a+b=2} R_X^a \boxtimes R_X^b.$$

We need an equality of Chow classes and a consistent numerical choice of the coefficient of Δ_X . We therefore give the recursion with the two output factors left unintegrated.

The reason for retaining these factors is already visible on X^3 . Let $p_{12} : X^3 \rightarrow X^2$ retain the first two factors, let p_i be the individual projections, and write Δ_{ij} for the locus where coordinates i and j agree. For $\gamma, \delta \in R_X$ homogeneous with total codimension two,

$$\begin{aligned}p_{12*}([\Delta_{13}] \cdot [\Delta_{23}]) &= [\Delta_X], \\ p_{12*}([\Delta_{13}] \cdot p_3^* \gamma \cdot p_2^* \delta) &= \gamma \boxtimes \delta.\end{aligned} \quad (5.7)$$

In the first identity both retained coordinates equal the third; the small diagonal maps isomorphically to Δ_X . In the second, only the first coordinate is tied to the third; the second remains independent. These are Chow identities, before any integration over the outputs. The Hilbert-scheme recursion below produces products of precisely such equality diagonals, together with line and tangent Chern classes. Their intersections will give the same two alternatives, including the excess factors when an equality is repeated.

Lemma 5.1 (Consecutive Hilbert-scheme recursion). *Let $r > 0$, and let $N_r = X^{[r-1, r]}$ be the scheme of nested subschemes $Z' \subset Z$ of lengths $r-1, r$. Write*

$$\psi : N_r \longrightarrow X^{[r]}, \quad \phi : N_r \longrightarrow X^{[r-1]}, \quad \rho : N_r \longrightarrow X$$

for the two forgetful maps and the residual-point map. Then N_r is smooth, projective, irreducible of dimension $2r$; $\psi_[N_r] = r[X^{[r]}]$; and, with quotient-line convention,*

$$\sigma = (\phi, \rho) : N_r = \mathbb{P}(\mathcal{I}_{r-1}) \longrightarrow X^{[r-1]} \times X.$$

Put $\mathcal{L} = \mathcal{O}_{N_r}(1)$ and $\zeta = c_1(\mathcal{L})$. If $j : N_r \hookrightarrow N_r \times X$ is the graph of ρ , then

$$0 \longrightarrow (\psi \times \text{id}_X)^* \mathcal{I}_r \longrightarrow (\phi \times \text{id}_X)^* \mathcal{I}_{r-1} \longrightarrow j_* \mathcal{L} \longrightarrow 0. \quad (5.8)$$

For a two-term vector bundle resolution $0 \rightarrow U_1 \rightarrow U_0 \rightarrow \mathcal{I}_{r-1} \rightarrow 0$ on $X^{[r-1]} \times X$, one has, for every $k \geq 0$,

$$\sigma_*(\zeta^k \cap [N_r]) = \left[\frac{c(U_1^\vee)}{c(U_0^\vee)} \right]_k \cap [X^{[r-1]} \times X]. \quad (5.9)$$

All these identities remain valid after adjoining other Hilbert or surface factors.

Proof. The geometric description and universal sequence are the consecutive nested-scheme construction of [EGL01, Section 1, equations (3)–(6)]. A quotient $\mathcal{I}_{Z'} \rightarrow \mathbb{C}_x$ determines the extra length-one subscheme and conversely. In families its quotient line is $\mathcal{L}$, giving the projectivization and (5.8). The smoothness and dimension of the consecutive nested scheme are the surface Hilbert-scheme facts recorded there. Over the open locus of r distinct points, forgetting which point is residual has degree r , proving the pushforward formula for ψ . The universal ideals are flat over their Hilbert bases, so their ordinary pullbacks in this sequence agree with their derived pullbacks.

Here is also the pushdown calculation, to fix its convention and sign. The universal ideal has projective dimension at most one: it is flat over the smooth Hilbert base and its surface fibers are torsion-free ideals. A sufficiently negative sum of ample line bundles surjects onto it; the kernel is locally free, by the fiberwise projective dimension bound and the local criterion of flatness. This gives the displayed global resolution. Write $a = \text{rk } U_1$, so $\text{rk } U_0 = a + 1$, and let $p : \mathbb{P}(U_0) \rightarrow X^{[r-1]} \times X$. The inclusion $N_r \subset \mathbb{P}(U_0)$ is the zero locus of the section of $p^* U_1^\vee \otimes \mathcal{O}(1)$ induced by $U_1 \rightarrow U_0$. Its codimension is a , because both N_r and the base of σ have dimension $2r$. The ambient space is smooth, so the section is regular and its top Chern class represents $[N_r]$.

For $\xi = c_1(\mathcal{O}_{\mathbb{P}(U_0)}(1))$ the quotient convention gives $p_*(\xi^{a+k}) = [c(U_0^\vee)^{-1}]_k$ for $k \geq 0$, and zero for smaller exponents. Hence the left side of (5.9) equals

$$\begin{aligned} p_*(\xi^k c_a(p^* U_1^\vee \otimes \mathcal{O}(1))) &= \sum_{b=0}^{\min(a, k)} c_b(U_1^\vee) [c(U_0^\vee)^{-1}]_{k-b} \\ &= \left[\frac{c(U_1^\vee)}{c(U_0^\vee)} \right]_k. \end{aligned}$$

This is also [EGL01, Section 1, Lemma 1.1]. In particular, for $r = 2$ the map σ is the blowup of $X \times X$ along its diagonal, $\zeta = -E$ for its exceptional divisor, and the formula gives $\sigma_*(\zeta^2) = -[\Delta_X]$. Flat base change and the projection formula give all the assertions with extra factors. $\square$

Lemma 5.2 (Reduction to decorated equality diagonals). *Fix a finite list of line bundles on X . Consider a finite polynomial of Chern characters of universal point ideals and their derived duals, together with Chern classes of these lines and of T_X pulled from the surface slots and equality diagonals between those slots, on*

$$X^{[n_1]} \times X^{[n_2]} \times X^\ell, \quad \ell \geq 2.$$

Keep two specified surface factors as outputs and push forward over the Hilbert factors and all other surface factors. The resulting Chow class is a finite rational linear combination of pushforwards of products of equality diagonals and line and tangent classes on powers of X . The coefficients and the list of such products can be chosen by one recursion independent of X and of the chosen line bundles.

Proof. We induct on $n_1 + n_2$. When this sum is positive, choose a positive length r in a fixed order of the Hilbert factors and pull back along ψ of Lemma 5.1, with all other factors unchanged. All spaces used here are smooth, so Chow pullbacks are defined. Projection formula and $\psi_*[N_r] = r[X^{[r]}]$ express the original pushforward as $1/r$ times the pushforward of this pullback.

In every surface slot, replace the pulled-back ideal by (5.8) in K -theory. Its extra term is the graph of the new residual point, with the line twist $\mathcal{L}$. For that graph, Grothendieck–Riemann–Roch gives

$$\mathrm{ch}(j_*\mathcal{L}) = j_*(e^\zeta \rho^* \mathrm{td}(T_X)^{-1}). \quad (5.10)$$

Indeed the graph is a regular embedding of codimension two with normal bundle ρ^*T_X . It is the pullback of the diagonal under $(\rho, \mathrm{id}_X) : N_r \times X \rightarrow X \times X$. This square is Tor independent: the two local equations of the diagonal cut out the graph regularly, in the same codimension. Thus the graph term in (5.10) is a polynomial in ζ , the equality diagonal between the new point slot and the slot in use, and tangent classes. For a perfect complex, the degree- k Chern character of its derived dual is $(-1)^k \mathrm{ch}_k$; derived duals therefore introduce no new kind of class.

After this substitution the integrand is a polynomial in ζ whose coefficients pull back from $X^{[r-1]} \times X$ and the unchanged factors. Apply (5.9) to each power of ζ . Its right side is a universal polynomial in the Chern classes of $\mathcal{I}_{r-1}$, equivalently in its Chern characters with rational coefficients. The Hilbert length has decreased by one; the added surface factor records the residual point. The two output factors have not been pushed forward at any step. Repeating the construction ends with Hilbert schemes of length zero, whose ideals are the structure sheaf. Only surface factors, their diagonals, and the specified decorations remain. The choices of Hilbert-factor order, slot order, and polynomial expansion order can be fixed once for all inputs. This proves the assertion. We have adapted the geometric recursion underlying the integral formula of [EGL01, Section 3, Proposition 3.1]: here the two output factors remain unintegrated throughout a sequence of Chow identities. $\square$

The recursion has eliminated the Hilbert factors without integrating the outputs. We now apply it to the localization integrand and use the remaining diagonal intersections to select its coefficient of Δ_X by a rule independent of the surface.

Proposition 5.3 (Chow universality of a fiber contribution). *The class in (5.6) admits a decomposition*

$$C_*(T, H; n_1, n_2) = c_*[\Delta_X] + \Gamma_*, \quad \Gamma_* \in \sum_{a+b=2} R_X^a \boxtimes R_X^b. \quad (5.11)$$

One can choose c_* by a fixed universal rule. Besides the integer parameters and the chosen weights, this rule depends only on

$$\int_X c_2(T_X), \quad \int_X UV \quad (U, V \in \{K, L_1, L_2, T, H\}). \quad (5.12)$$

It gives $c_* = 0$ when $n_1 = n_2 = 0$. The same rule applies on every smooth projective surface with the same data and $\chi(\mathcal{O}_X) = 1$, whether or not the lines T, H are effective; effectivity is needed only for realizing the formula as a fixed contribution.

Proof. We first put the integrand into the form required by Lemma 5.2. For the surface projection $\pi_X : M \times X \rightarrow M$, Grothendieck–Riemann–Roch gives

$$\mathrm{ch}(\mathrm{RHom}(\mathcal{I}_i, \mathcal{I}_j(P))) = \pi_{X*}(\mathrm{ch}(\mathcal{I}_i^\vee) \mathrm{ch}(\mathcal{I}_j) e^P \mathrm{td}(T_X)), \quad (5.13)$$

$$\mathrm{ch}(P^{[n_i]}) = \pi_{X*}((1 - \mathrm{ch}(\mathcal{I}_i)) e^P \mathrm{td}(T_X)). \quad (5.14)$$

The dual in (5.13) is the derived dual. These identities hold in Chow, since the ideals are perfect and projection is smooth and proper. The constant complex $R\Gamma(P)$ has K -class $\chi(P)[\mathcal{O}_M]$, with $\chi(P) = 1 + P(P - K)/2$. The Chern classes in the Euler factors are universal polynomials in the resulting Chern characters. The incidence insertion is $[Z_i] = -\mathrm{ch}_2(\mathcal{I}_i)$. Thus, after the finite truncation required by (5.6), every term is a polynomial in the classes of the preceding Lemma and in pushforwards of such classes.

Products of surface pushforwards introduce only additional surface slots. More precisely, if a, b are classes on $M \times X$, the projection formula and flat base change identify $\pi_{X*}(a)\pi_{X*}(b)$ with the pushforward of $a_{(1)}b_{(2)}$ from $M \times X \times X$. The same statement holds over the base M together with the two marked factors and can be iterated for any finite product. Apply Lemma 5.2 to the resulting expressions.

It remains to perform the surface pushforwards. This reduction will both prove the Chow decomposition and specify the coefficient to be compared across surfaces. Represent a monomial in equality diagonals by a graph $\mathcal{G}$: its vertices are the surface slots and each factor $[\Delta_{uv}]$ is an edge between slots u, v , with repeated factors recorded as repeated edges. For each connected component C , let v_C, e_C count its vertices and edges. Let

$$\iota : \prod_{C \text{ component of } \mathcal{G}} X \longrightarrow X^{\{\text{vertices of } \mathcal{G}\}}$$

assign the same coordinate to all vertices in a component, and let pr_C project its domain to the factor indexed by C . Then

$$\prod_{uv \text{ edge of } \mathcal{G}} [\Delta_{uv}] = \iota_* \left(\prod_{C \text{ component of } \mathcal{G}} (\mathrm{pr}_C^* c_2(T_X))^{e_C - v_C + 1} \right). \quad (5.15)$$

Indeed, a spanning tree in C imposes $v_C - 1$ transverse equalities, giving its small diagonal. Each of the remaining $e_C - v_C + 1$ edges imposes an equality already satisfied. The diagonal self-intersection formula contributes $c_2(T_X)$ for each such edge. The exponent is zero for an isolated vertex as well as for a tree. By projection formula, every line or tangent decoration is then restricted to the copy of X belonging to its component.

There are three possibilities for the subsequent pushforward:

- (i) A component containing neither output is integrated over X . It contributes zero unless its decoration has codimension two; otherwise it contributes the intersection number of that decoration.

- (ii) A component containing exactly one output maps isomorphically to that output and leaves its decoration there. When the outputs belong to different components, their decorations therefore give an external product of total codimension two, multiplied by the intersection numbers from components of type (i).
- (iii) A component containing both outputs maps to their diagonal. It contributes $\Delta_{X*}(\eta)$, multiplied by the same kind of intersection numbers. Since Δ_X already has codimension two, the codimension-two part of the answer uses only $\eta \in \text{CH}^0(X) = \mathbb{Q}[X]$. Thus this case gives a scalar multiple of $[\Delta_X]$.

Fix the recursion and edge-processing orders once for all inputs. Carry out this reduction on the formal decorated graphs, replacing components of type (i) by formal intersection numbers. Define c_* as the sum of the scalar coefficients in case (iii), and assign the terms with separated outputs to Γ_* . Only after this assignment do we evaluate the intersection numbers and Chow classes on X . Equation (5.15) and the three cases prove (5.11) in Chow and specify the promised choice of its coefficient.

Every codimension-two decoration integrated over a surface is a linear combination of $c_2(T_X)$ and products of the lines in (5.12), since $c_1(T_X) = -K$. The scalar Euler exponents are determined by the same data through Riemann–Roch. Hence the rule gives the same c_* on any two surfaces with those numerical data. We do not require the resulting diagonal decomposition to be unique on an individual surface: relations between its diagonal and external products are never used to select c_* .

Finally, when both point lengths are zero, M is a point, $\mathcal{E} = 1$, and $E_{ij} = 0$. There is no incidence term in J . The two copies of J involve only classes on their separate output factors and the scalar variables x_i, w_i . The graph procedure has no edge joining the outputs, so it selects $c_* = 0$. None of this formal calculation requires sections of T or H , proving the last assertion as well. $\square$

All diagonal coefficients below use this fixed formal choice. In particular, numerical comparison transports the coefficient of the explicit fiber formula (5.6); it does not transport the geometry or multiplicity of the divisor component from which that formula arose.

5.4 The relation, cohomological tests, and its action

The fixed contributions now have Chow decompositions with consistently chosen diagonal coefficients. Summing them gives the required relation; cohomological pairings will then test its coefficient, and its correspondence action will control zero-cycles.

Theorem 5.4 (Chow relation). *For either surface setting, and any $p \geq 0$ with $v > p + 2$, sum the classes (5.6) with the following multiplicities:*

- (i) *If $p_g = q = 0$, sum over the effective ordered line pairs $T + H = D$ and nonnegative point lengths satisfying (4.2), with $m_T, m_H \geq 0$. Each fixed part has multiplier 1.*
- (ii) *In the auxiliary setting, $D = 3K$. Sum over the ordered pairs $(3K, 0), (0, 3K)$ with multiplier 1, and $(2K, K), (K, 2K)$ with multiplier $-s$, with the admissible point lengths. Here $s > 0$ is the canonical-divisor coefficient of (4.16).*

All sums are finite. Writing c for the same weighted sum of the coefficients chosen in Proposition 5.3, one obtains

$$0 = c[\Delta_X] + \Gamma \quad \text{in } \text{CH}^2(X \times X), \quad (5.16)$$

where Γ is a sum of external products of total codimension two.

Proof. The finite-type fixed locus has only finitely many discrete decompositions and Hilbert polynomials, as in Section 4. Proposition 4.4 gives the first list and discards negative m_i . Proposition 4.9 gives the second list, its fiber restrictions, and its multiplicities. The moving factor and insertions were computed in (5.4) and (5.5), so these are exactly the contributions to the ordinary codimension-two part of localization of (5.1). That part vanishes by (5.2). Substitute (5.11) for each contribution to obtain (5.16). $\square$

For the subsequent nonvanishing arguments, let $\alpha \in H^2(X, \mathbb{R})$ and write $\alpha U = \int_X \alpha \smile U$ and $\alpha^2 = \int_X \alpha \smile \alpha$. Recall $\pi_X : M \times X \rightarrow M$ and set

$$\mu_i(\alpha) = \pi_{X*}([Z_i] \smile \alpha), \quad \mu_\alpha = \pi_{X*}(J \smile \alpha).$$

Here $[Z_i]$ and J denote their cycle classes; α need not be algebraic. Expansion of (5.5) gives the useful formula

$$\mu_\alpha = \sum_{i=1}^2 (\mu_i(\alpha) + (w_i - x_i)(\alpha D_i) + x_i(\alpha L_i)). \quad (5.17)$$

Integrating over the two output factors first and using projection formula shows that

$$\begin{aligned} \int_{X \times X} C_*(T, H; n_1, n_2) \smile (\alpha \boxtimes \alpha) \\ = [x_1^{m_T} x_2^{m_H}] \int_M \mathcal{E}(x) \mathcal{R}(x) (x_1 + x_2)^p \mu_\alpha^2. \end{aligned} \quad (5.18)$$

The integral takes the top ordinary cohomological degree on M . It automatically selects precisely the ordinary codimension-two part on the left. In particular, no algebraicity assumption on α is used.

If α is orthogonal to K, L_1, L_2, T, H , then

$$\int_{X \times X} C_*(T, H; n_1, n_2) \smile (\alpha \boxtimes \alpha) = c_* \alpha^2. \quad (5.19)$$

Indeed, an external product of codimensions $(0, 2)$ or $(2, 0)$ pairs to zero by degree, and one of codimensions $(1, 1)$ pairs to the product of two intersections with α . Those intersections vanish because the codimension-one part of R_X is spanned by the listed line classes. The diagonal pairs to α^2 . This is a test for the chosen coefficient of a relation already proved in Chow.

Corollary 5.5. *If $p_g = q = 0$ and the coefficient c in (5.16) is nonzero, then $\mathrm{CH}_0(X)^0 \otimes \mathbb{Q} = 0$.*

Proof. For $Z \in \mathrm{CH}^2(X \times X)$, use its correspondence action

$$Z_*(z) = \mathrm{pr}_{2*}(\mathrm{pr}_1^* z \cdot Z).$$

The diagonal acts as the identity. If $Z = a \boxtimes b$ has total codimension two and a has positive codimension, then $z \cdot a = 0$ on the surface for every zero-cycle z , by dimension. If a has codimension zero, connectedness of X gives $a = \lambda[X]$ and $Z_*(z) = \lambda \deg(z)b$. Every summand of Γ therefore annihilates degree-zero zero-cycles. Acting by (5.16) gives $cz = 0$ in $\mathrm{CH}_0(X) \otimes \mathbb{Q}$. Since $c \neq 0$, this implies $z = 0$. $\square$

6 Nonvanishing when the minimal intersection lattice has rank at least two

Let X_0 be a minimal surface of general type with $p_g = q = 0$, and write $K_0 = K_{X_0}$ and $k_0 = K_0^2$. By Proposition 2.2, K_0 is nef and big and $1 \leq k_0 \leq 9$. This Section treats $k_0 \leq 8$.

Proposition 6.1. *Suppose $1 \leq k_0 \leq 8$. Blow up*

$$e = \max\{k_0 - 5, 0\}$$

distinct points of X_0 , and call the resulting surface X . There are choices of L_1, L_2, D, n, p on X for which Theorem 5.4 gives a relation with $c > 0$. Consequently the degree map on $\text{CH}_0(X)_\mathbb{Q}$ is an isomorphism.

The proof pairs the Chow relation with $\alpha \boxtimes \alpha$, for a class $\alpha \in H^2(X, \mathbb{R})$ with $\alpha K_X = 0$ and $\alpha^2 < 0$. We will take L_1, L_2, D to be integral multiples of K_X . A fixed contribution indexed by $T + H = D$ with $\alpha H = 0$ then pairs to $c_* \alpha^2$, since its external products pair to zero. We arrange that every remaining contribution has zero point lengths, hence $c_* = 0$, and has positive pairing. At least one such contribution will occur, so the vanishing of the full pairing will force $c > 0$. The lattice calculation supplies α ; the subsequent numerical bounds isolate the contributions with zero point lengths.

6.1 A characteristic-vector calculation

The exponential sequence and $H^1(\mathcal{O}_{X_0}) = H^2(\mathcal{O}_{X_0}) = 0$ identify $\text{Pic}(X_0)$ with $H^2(X_0, \mathbb{Z})$. In particular every integral cohomology class is the first Chern class of a line bundle. The free quotient

$$\Lambda = H^2(X_0, \mathbb{Z})/\text{torsion}$$

is unimodular by Poincaré duality. Its signature is $(1, b)$ with $b = 9 - k_0$, by Hodge index and $b_2(X_0) = 10 - k_0$. Riemann–Roch gives

$$x^2 \equiv K_0 x \pmod{2} \quad (x \in \Lambda),$$

so K_0 is a characteristic vector.

We use the classification of indefinite unimodular integral forms [Ser73, Chapter V, §2.1, Corollary 1 to Theorem 2, and §2.2, Theorems 4–6]: an odd form of signature $(1, b)$ is $I_{1,b} = \langle 1 \rangle \oplus \langle -1 \rangle^{\oplus b}$; the signature difference of an even unimodular form is divisible by eight. Since $1 \leq b \leq 8$, the even case has $b = 1$ and is the hyperbolic plane.

Lemma 6.2. *If Λ is odd, there is a line bundle h_0 on X_0 with*

$$h_0^2 = -1, \quad K_0 h_0 = 1.$$

If Λ is even, then $k_0 = 8$ and there is a line bundle h_0 with

$$h_0^2 = 0, \quad K_0 h_0 = 2.$$

Proof. In the odd case choose diagonal coordinates and write $K_0 = (u; v_1, \dots, v_b)$. The characteristic property says that all these integers are odd. Coordinate sign changes and permutations allow us to assume

$$u > 0, \quad v_1 \geq \dots \geq v_b > 0, \quad u^2 = 9 - b + \sum_{i=1}^b v_i^2.$$

If $v_i = 1$, the class $-e_i$, for the corresponding negative basis vector e_i , has square -1 and pairing 1 with K_0 . We show by descent on u that this case can always be reached by integral lattice isometries.

For $b = 1$, the equation $(u - v_1)(u + v_1) = 8$ forces $(u, v_1) = (3, 1)$: both factors are positive even integers and the second is strictly larger than the first. Suppose next that $b \geq 3$ and that every $v_i \geq 3$. Put $\ell = v_1 + v_2 + v_3$. The ordering of the v_i gives

$$\begin{aligned}\ell^2 - u^2 &= 2(v_1v_2 + v_1v_3 + v_2v_3) - \sum_{i=4}^b v_i^2 - (9 - b) \\ &\geq (9 - b)(v_3^2 - 1) > 0.\end{aligned}$$

On the other hand, $\ell^2 \leq 3(v_1^2 + v_2^2 + v_3^2) < 3u^2 < 4u^2$. The vector $z = (1; 1, 1, 1, 0, \dots, 0)$ has square -2 . Its integral reflection $x \mapsto x + (xz)z$ changes the positive coordinate of K_0 to

$$u' = 2u - \ell, \quad 0 < u' < u.$$

For $b = 2$ and $v_1, v_2 \geq 3$, put $\ell = v_1 + v_2$. Here

$$\ell^2 - u^2 = 2v_1v_2 - 7 > 0, \quad \ell^2 \leq 2(u^2 - 7) < 2u^2 < \frac{9}{4}u^2.$$

The vector $z = (1; 1, 1)$ has square -1 , so its reflection is the integral isometry $x \mapsto x + 2(xz)z$. It changes the positive coordinate to

$$u' = 3u - 2\ell, \quad 0 < u' < u.$$

In either case the reflected vector remains characteristic and of square $9 - b$. Rechoose the signs and order of its negative coordinates and repeat. The positive integer u strictly decreases, so the process must reach a coordinate $v_i = 1$. Pulling the corresponding vector back through the isometries gives the asserted class, which lifts to a line bundle.

In the even case choose an isotropic basis f, g with $fg = 1$. Writing $K_0 = af + bg$, the characteristic property makes a, b even, and $2ab = K_0^2 = 8$. After changing both basis signs if necessary, $a = b = 2$. Either f or g gives h_0 . $\square$

Let $\pi : X \rightarrow X_0$ be the blowup in Proposition 6.1, with exceptional curves $E_1, \dots, E_e$. We usually suppress π^* on line and cohomology classes. Thus

$$K = K_X = \pi^*K_0 + \sum_{j=1}^e E_j, \quad k = K^2 = k_0 - e = \min\{k_0, 5\} > 0.$$

In the odd case put $h' = \pi^*h_0$. In the even case $k_0 = 8$ and $e = 3$, so we may put $h' = \pi^*h_0 + E_1$. Both choices satisfy

$$(h')^2 = -1, \quad Kh' = 1. \tag{6.1}$$

The symbol h' denotes a line bundle and is distinct from the insertion class h on the Quot scheme.

Set $a = K_0h_0$, so $a = 1$ in the odd case and $a = 2$ in the even case, and define

$$\alpha = \pi^*\left(h_0 - \frac{a}{k_0}K_0\right) \in H^2(X, \mathbb{R}). \tag{6.2}$$

Pullback classes are orthogonal to the exceptional curves, whence

$$\alpha K = 0, \quad \alpha h' = \alpha^2 = h_0^2 - \frac{a^2}{k_0} < 0. \tag{6.3}$$

6.2 Positive canonical pairings on the blowup

The following bound will apply to each of the two effective divisor classes in a fixed component.

Lemma 6.3. *Let P be an effective line bundle on X such that $m_P \geq 0$ and $\alpha P \neq 0$. Then $KP > k$.*

Proof. Write

$$P = \pi^* P_0 + \sum_{j=1}^e a_j E_j, \quad r_0 = K_0 P_0,$$

where the a_j are integers. Pushing forward an effective divisor representing P shows that P_0 is effective; consequently $r_0 \geq 0$ by nefness of K_0 . The inequality $2m_P = P^2 - KP \geq 0$ gives

$$0 \leq \sum_{j=1}^e a_j(a_j - 1) \leq P_0^2 - r_0 < \frac{r_0^2}{k_0} - r_0. \quad (6.4)$$

The first inequality holds for all integral a_j , including negative ones. The last is strict Hodge index: $\alpha P \neq 0$ implies that P_0 is not numerically proportional to K_0 . Together with $r_0 \geq 0$, this forces $r_0 > k_0$.

We next prove $KP > 0$. If $e = 0$, this is immediate. Otherwise put $A_0 = \sum_j a_j$ and suppose $KP = r_0 - A_0 \leq 0$. By Cauchy–Schwarz,

$$\sum_j a_j(a_j - 1) \geq \frac{A_0^2}{e} - A_0 \geq \frac{r_0^2}{e} - r_0 > \frac{r_0^2}{k_0} - r_0.$$

Indeed $A_0 \geq r_0 > k_0 > e$, and $u \mapsto u^2/e - u$ is increasing on $[r_0, \infty)$. This contradicts (6.4). Thus $KP > 0$.

Since $K^2 = k > 0$, the intersection form is negative definite on $K^\perp$. Also $\alpha P \neq 0$ and $\alpha K = 0$ show that P is not numerically proportional to K . Therefore

$$0 < KP \leq P^2 < \frac{(KP)^2}{k},$$

where the middle inequality is $m_P \geq 0$. Dividing by $KP > 0$ proves $KP > k$. This application of Hodge index requires the positive square of K ; all effectivity bounds above used the nef class $\pi^* K_0$. $\square$

6.3 All contributions with nonzero divisor pairing

We choose the Quot data so that every fixed contribution detected by the divisor pairing with α has zero point lengths. This will make those contributions explicit, while the orthogonal terms account for the diagonal coefficient. Set

$$\begin{array}{c|cc} & d & p \\ \hline k=1 & 4 & 0 \\ 2 \leq k \leq 5 & 3 & k-2 \end{array} \quad D = dK, \quad L_1 = 0, \quad L_2 = K, \quad n = d(k+1). \quad (6.5)$$

Here $A = -K$, so (3.7) gives $v = 2d$. For $k = 1$ we have $v = 8 > 2 = p + 2$; for $2 \leq k \leq 5$ we have $v = 6 > k = p + 2$. Thus Theorem 5.4 applies. This degree inequality is the reason for using blowups to reduce k to at most five.

By Proposition 4.4, only effective line decompositions $T + H = D$ with $m_T, m_H \geq 0$ need be retained. Write

$$r = KH, \quad s = H^2, \quad N = n_1 + n_2.$$

For these data the length constraint (4.2) is

$$N = d(k+1) - (d+1)r + s \geq 0. \quad (6.6)$$

Lemma 6.4. *Every such fixed contribution with $\alpha H \neq 0$ satisfies*

$$KH = H^2 = k+1, \quad n_1 = n_2 = 0, \quad m_H = 0, \quad m_T = p. \quad (6.7)$$

Proof. Since $\alpha T = -\alpha H \neq 0$, Lemma 6.3 applies to both H and T . It gives

$$r > k, \quad dk - r > k, \quad r \leq s < \frac{r^2}{k}.$$

Furthermore $s - r$ is even by Riemann–Roch.

If $k = 1$, then $d = 4$ and the first two bounds force $r = 2$. The inequalities $2 \leq s < 4$ and $s \equiv 2 \pmod{2}$ force $s = 2$. Equation (6.6) gives $N = 0$.

For $2 \leq k \leq 5$, write $r = k + j$. Since $d = 3$, the two strict canonical bounds give $1 \leq j \leq k - 1$. Use the parity condition by writing $s = r + 2m$, where $m = m_H$ is a nonnegative integer. The strict Hodge bound gives

$$2m < \frac{r^2}{k} - r = j + \frac{j^2}{k} < 2j. \quad (6.8)$$

Consequently $m \leq j - 1$. Substituting into (6.6) yields

$$0 \leq N = 3(1 - j) + 2m \leq 1 - j. \quad (6.9)$$

It follows that $j = 1$, then $m = 0$ and $N = 0$, so $r = s = k + 1$. This integral bound excludes every $j \geq 2$, including the boundary values for $k = 3$ and $k = 4$, without a separate numerical enumeration.

In every case $r = s = k + 1$ and $N = 0$. Since $n_1, n_2 \geq 0$, they both vanish. Finally

$$2m_T = (dK - H)^2 - K(dK - H) = d(d-1)k - 2(d-1)(k+1),$$

which equals 0 for $(k, d) = (1, 4)$ and $2(k-2)$ for $d = 3$. This proves all assertions. $\square$

For reference, the possibilities proved above are

k	d	p	KH	H^2	$n_1 + n_2$	m_T	m_H
1	4	0	2	2	0	0	0
2	3	0	3	3	0	0	0
3	3	1	4	4	0	1	0
4	3	2	5	5	0	2	0
5	3	3	6	6	0	3	0

(6.10)

6.4 An effective pair and the sign of its contribution

Lemma 6.5. *The actual line bundles*

$$H = K + h', \quad T = (d-1)K - h'$$

are effective and satisfy (6.7), with $\alpha H \neq 0$.

Proof. Equation (6.1) gives $KH = H^2 = k+1$. The calculations in the preceding proof then give $m_H = 0$, $m_T = p$, and $N = 0$. Consequently

$$\chi(H) = 1, \quad \chi(T) = p+1 > 0.$$

Let $B_0 = \pi^*K_0$, a nef divisor on X . Its pairings with the Serre-dual line bundles are

$$B_0(K-H) = -a < 0, \quad B_0(K-T) = (2-d)k_0 + a < 0.$$

For the second inequality, in the odd case its value is -1 when $k_0 = 1$ and $1 - k_0 < 0$ when $k_0 \geq 2$; in the even case it is $-8 + 2 = -6$. Thus neither $K-H$ nor $K-T$ has a section, since an effective divisor has nonnegative intersection with a nef divisor. Serre duality gives $H^2(H) = H^2(T) = 0$, and Riemann–Roch now yields

$$h^0(H) = \chi(H) + h^1(H) > 0, \quad h^0(T) = \chi(T) + h^1(T) > 0.$$

Both line bundles are therefore effective. Finally $\alpha H = \alpha h' = \alpha^2 < 0$ by (6.3). Thus this pair occurs in the fixed locus with both point subschemes empty, by Lemma 4.1. $\square$

Proof of Proposition 6.1. Use the data (6.5) and pair the Chow relation of Theorem 5.4 with $\alpha \boxtimes \alpha$ in cohomology. We compute the contributions using (5.6) and (5.18).

For any contribution with $\alpha H = 0$, the class α is orthogonal to K, L_1, L_2, T, H . Thus its external-product part has zero pairing, and it contributes $c_*\alpha^2$ by Proposition 5.3.

For every remaining contribution, Lemma 6.4 gives $n_1 = n_2 = 0$. Its chosen diagonal coefficient c_* is therefore zero by Proposition 5.3. Its point Hilbert factors and fixed Euler factors are trivial. Formula (5.5) gives

$$\mu_\alpha = (\alpha H)(-w_1 + w_2 + x_1 - x_2) = (\alpha H)(-1 + x_1 - x_2).$$

Since $m_T = p$ and $m_H = 0$, its full pairing is

$$\begin{aligned} [x_1^p x_2^0] \mathcal{R}(x_1, x_2) (x_1 + x_2)^p (\alpha H)^2 (-1 + x_1 - x_2)^2 \\ = (\alpha H)^2 \mathcal{R}(0, 0). \end{aligned}$$

Here all denominators in $\mathcal{R}$ have nonzero constant term. Setting $x_2 = 0$ and extracting x_1^p therefore uses only the constant term of the factors multiplying x_1^p .

We compute this constant exactly. Since there are no point factors, (5.4) reduces at $(x_1, x_2) = (0, 0)$ to the scalar weight powers. The $i = 2, j = 1$ weight is 1, and the $i = 1, j = 2$ weight is -1 , with

$$P_{12} = T + K, \quad B_{12} = T - H + K.$$

Thus

$$\begin{aligned} \mathcal{R}(0, 0) &= (-1)^{\chi(T-H+K) - \chi(T+K)} \\ &= (-1)^{-TH + (H^2 - KH)/2} = (-1)^{-TH}. \end{aligned}$$

The second equality follows by expanding Riemann–Roch, and the last uses $H^2 = KH$. Moreover

$$TH = (dK - H)H = (d - 1)(k + 1) = \begin{cases} 6, & k = 1, \\ 2(k + 1), & 2 \leq k \leq 5, \end{cases}$$

which is even. Hence every such term has

$$\mathcal{R}(0, 0) = 1. \quad (6.11)$$

Let the sum below run over the effective ordered line decompositions occurring in the fixed locus with $\alpha H \neq 0$, retaining distinct torsion twists as distinct terms. This is a finite sum, as the fixed locus has only finitely many component data. It is nonempty by Lemma 6.5. Summing the two kinds of contributions therefore gives

$$0 = c\alpha^2 + \sum_{\alpha H \neq 0} (\alpha H)^2, \quad c = -\frac{\sum_{\alpha H \neq 0} (\alpha H)^2}{\alpha^2} > 0. \quad (6.12)$$

In this equality c is the coefficient already chosen in the Chow relation: the second group of terms has $c_* = 0$, and the first group accounts for its entire sum. The strict positivity follows from $\alpha^2 < 0$ and the nonempty sum of positive real squares.

Corollary 5.5 now gives the rational degree isomorphism on X . Lemma 2.1 passes it to X_0 , proving the Proposition. $\square$

7 The case $K^2 = 9$

We now treat a minimal surface with $p_g = q = 0$ and $K^2 = 9$, whose intersection lattice has rank one. The argument will reduce its diagonal coefficient to two numerical divisor types. These types occur with nonnegative multiplicities on the given surface. On an auxiliary surface with the same Chern numbers, the canonical-divisor type instead has a negative virtual multiplicity. Comparing the two relations will leave just one coefficient to evaluate, on a Hilbert scheme of length two.

Throughout this Section, c_* means the coefficient chosen by the fixed recursion of Proposition 5.3. Its numerical selection rule is the same on both surfaces; uniqueness of a diagonal decomposition on either surface is not required.

Proposition 7.1. *Let X be a smooth minimal projective complex surface of general type with $p_g(X) = q(X) = 0$ and $K_X^2 = 9$. Take*

$$L_1 = L_2 = 0, \quad D = 3K_X, \quad n = 20, \quad p = 7. \quad (7.1)$$

The coefficient c in the Chow relation of Theorem 5.4 is strictly positive.

7.1 Effective line classes and their multiplicities

Write $K = K_X$. The reductions in Section 2 give $b_2(X) = 1$ and $\int_X c_2(T_X) = 3$. The exponential sequence, using $H^1(X, \mathcal{O}_X) = H^2(X, \mathcal{O}_X) = 0$, identifies $\text{Pic}(X)$ with $H^2(X, \mathbb{Z})$. Its quotient by torsion has a positive unimodular intersection pairing of rank one. Choose its generator G on the ray of an ample divisor. Then

$$G^2 = 1, \quad K \equiv 3G,$$

where $\equiv$ denotes numerical equivalence. Indeed $K^2 = 9$ gives the coefficient ± 3 , and the nef and big canonical class has positive intersection with an ample divisor. Every irreducible curve has a positive coefficient on the same ray: its intersection with an ample divisor is positive. Consequently K has positive intersection with every curve, and $K^2 > 0$. The Nakai–Moishezon criterion proves that K is ample.

For the data (7.1), Equation (3.7) gives

$$v = 2 \cdot 20 - K(3K) = 13 > 9 = p + 2.$$

Write $H \equiv aG$ and $T \equiv (9 - a)G$. Effectivity of both lines implies $0 \leq a \leq 9$. Their virtual divisor dimensions and total point length, by Equation (4.2), are

$$m_H = \frac{a(a-3)}{2}, \quad m_T = \frac{(9-a)(6-a)}{2}, \quad n_1 + n_2 = 20 - 9a + a^2. \quad (7.2)$$

Here and below an effective numerical class can represent several distinct line bundles; their contributions must all be counted. The complete numerical list is

a	0	1	2	3	4	5	6	7	8	9
m_H	0	-1	-1	0	2	5	9	14	20	27
m_T	27	20	14	9	5	2	0	-1	-1	0
$n_1 + n_2$	20	12	6	2	0	0	2	6	12	20

Proposition 4.4 eliminates $a = 1, 2, 7, 8$ because one of the two virtual divisor dimensions is negative. At $a = 4, 5$ the point lengths are both zero, so Proposition 5.3 gives $c_* = 0$. Only the outer numerical type $(3K, 0)$ and the middle numerical type $(2K, K)$, with both orders included, can contribute to c .

A numerically trivial effective line bundle is trivial: a nonzero effective divisor has positive intersection with an ample divisor, whereas a section with empty zero divisor trivializes its line bundle. Hence the outer type consists of the single unordered pair $(3K, 0)$. The lines numerically equivalent to K are precisely $K + \epsilon$, where ϵ runs over $\text{Pic}(X)_{\text{tors}}$. This torsion group is finite, since $H^2(X, \mathbb{Z})$ is finitely generated. The untwisted line K has no section. For $\epsilon \neq 0$, the preceding observation gives $H^0(X, -\epsilon) = 0$, and therefore

$$h^2(X, K + \epsilon) = 0, \quad \chi(K + \epsilon) = 1, \quad h^0(X, K + \epsilon) = 1 + h^1(X, K + \epsilon) > 0.$$

Every such line is effective. Its complement $2K - \epsilon$ is also effective: $(2K - \epsilon) - K$ is ample, so Kodaira vanishing and Riemann–Roch give $h^0(X, 2K - \epsilon) = 10$. The same argument gives $h^0(X, 3K) = 28$. Thus all the required high lines occur.

Define

$$\tau_X = \#(\text{Pic}(X)_{\text{tors}} \setminus \{0\}). \quad (7.3)$$

The notation τ_X keeps this count distinct from the equivariant parameter t . Let c_{out} be the sum of the universal coefficients for the two orders of the numerical pair $(3K, 0)$, including every distribution $n_1 + n_2 = 20$. Let c_{mid} be the analogous sum for the two orders of $(2K, K)$ and all distributions $n_1 + n_2 = 2$. These numbers are defined by Equation (5.6) and Proposition 5.3, including when the indicated precise line bundles are not effective on X . Their numerical inputs are $K^2 = 9$, $\int_X c_2(T_X) = 3$, $\chi(\mathcal{O}_X) = 1$, and (7.1).

For each nontrivial ϵ , the two orders of $(2K - \epsilon, K + \epsilon)$ have exactly these middle numerical inputs. Each unordered pair is counted once by ϵ , since its two members have different numerical classes. The linear systems need not have the same actual dimension as ϵ varies; their H^1 obstruction

factors have already been incorporated in the coefficient extraction of Proposition 4.4. It follows that the total coefficient on X is

$$c = c_{\text{out}} + \tau_X c_{\text{mid}}. \quad (7.4)$$

The outer sum involves Hilbert schemes of total length 20, whereas the middle sum has total length 2. We will use the auxiliary relation to express c_{out} as a positive multiple of c_{mid} , avoiding the length-20 calculation. Positivity will then follow from the explicit length-two computation.

7.2 Comparison on the Cartwright–Steger surface

Let Y be the Cartwright–Steger surface. Its existence as a smooth projective compact two-ball quotient follows from the torsion-free lattice in [CKY17, Theorem 2]. Its invariants are

$$K_Y^2 = 9, \quad \int_Y c_2(T_Y) = 3, \quad \chi(\mathcal{O}_Y) = 1, \quad q(Y) = p_g(Y) = 1; \quad (7.5)$$

see [CKY17, Main Theorem (a) and Lemma 3]. The Bergman metric on the ball descends to Y , and its negative Ricci form gives a positive curvature form on K_Y . Kodaira’s positivity criterion therefore makes K_Y ample. Thus Y satisfies all the hypotheses of the auxiliary construction of Proposition 4.9 with $D = 3K_Y$.

Choose a nonzero $\omega \in H^0(Y, K_Y)$ and set $\alpha = [\omega + \bar{\omega}] \in H^2(Y, \mathbb{R})$. Holomorphic top forms are closed, and

$$\alpha^2 := \int_Y \alpha \smile \alpha = 2 \int_Y \omega \wedge \bar{\omega} > 0. \quad (7.6)$$

The inequality follows in local holomorphic coordinates from positivity of the associated real volume form. Hodge type gives $\alpha \cdot P = 0$ for every divisor class P on Y .

Lemma 7.2. *For the common universal coefficients defined above, one has*

$$c_{\text{out}} - s c_{\text{mid}} = 0, \quad s > 0, \quad (7.7)$$

where $-s$ is the canonical-divisor coefficient in Equation (4.16) for Y .

Proof. Apply the construction on Y with the data (7.1). Proposition 4.9 first replaces the low divisor virtual class by the empty-divisor point with coefficient 1, or by the canonical-divisor point with coefficient $-s$. The compatible virtual pullback along the high morphism is its smooth pullback multiplied by the high point-obstruction Euler class. Thus the surviving terms are computed on

$$|3K_Y| \times Y^{[n_1]} \times Y^{[n_2]} \quad \text{or} \quad |2K_Y| \times Y^{[n_1]} \times Y^{[n_2]},$$

with total point lengths 20 and 2, respectively, and with both fixed point-obstruction Euler factors retained. The high linear systems have dimensions 27 and 9. The low divisor is fixed, so its hyperplane variable is zero. Both orders occur, and the canonical multiplier is $-s$ in each order.

The projection formula restricts all operational insertions to these actual fibers. Derived base change likewise restricts the moving complexes, giving exactly Equation (5.4). Consequently the remaining formulas are those defining c_{out} and c_{mid} , with their indicated multipliers. This restriction precedes the use of universality: no integration over a varying Picard component is being compared by numerical invariants.

After that restriction, the equality of the numerical data in (7.5) with those for X permits Proposition 5.3 to identify the chosen coefficients. The resulting Chow relation on Y^2 has diagonal coefficient $c_{\text{out}} - s c_{\text{mid}}$. Pair it with $\alpha \boxtimes \alpha$. Every external term vanishes: bidegree allows only a divisor on each factor to pair nontrivially, and each such divisor is orthogonal to α . The diagonal pairs to α^2 . Equation (7.6) now gives (7.7). $\square$

7.3 The middle coefficient

Lemma 7.3. *For the data (7.1), with numerical inputs $K^2 = 9$ and $\int c_2 = 3$, the two-order middle coefficient is*

$$c_{\text{mid}} = 12.$$

Proof. Compute on Y with the class α in (7.6), omitting the overall canonical multiplier $-s$. We compute the order $T = 2K_Y$, $H = K_Y$ first, and verify equality with the reverse order at the end. For this order,

$$n_1 + n_2 = 2, \quad m_T = 9, \quad m_H = 0.$$

The three possible point-length distributions are $(2, 0)$, $(1, 1)$, and $(0, 2)$. Taking the coefficient of x_2^0 means setting $x_2 = 0$: every moving weight in Equation (5.4) has nonzero constant term 1 or -1 after $t = 1/2$, so all its inverse Euler factors expand with nonnegative powers of x_1, x_2 . This remains true for virtual bundles, by expanding their Chern classes at the nonzero weight. If $n_2 > 0$, the low fixed factor at $x_2 = 0$ is

$$e((\mathcal{O}_Y^{[n_2]})^\vee) = 0.$$

Indeed the constant function 1 is a nowhere-zero section of the rank- n_2 tautological bundle, so its top Chern class, and hence that of its dual, vanishes. Equivalently its Euler polynomial is divisible by x_2 ; there is no negative power of x_2 in the other factors to cancel it. Therefore only $(n_1, n_2) = (2, 0)$ contributes.

Put

$$U = (-K_Y)^{[2]}, \quad N = \mathcal{O}_Y^{[2]}, \quad x = x_1, \quad \mu = \int_Y [Z_1] \alpha \in H^2(Y^{[2]}, \mathbb{R}).$$

All the other terms of Equation (5.5) integrate to zero against α , by its orthogonality to divisors and by degree. Consequently μ is the class used in the pairing formula (5.18).

For this order, the line data in the moving ratio are

$$P_{12} = 2K_Y, \quad P_{21} = K_Y, \quad B_{12} = K_Y, \quad B_{21} = -K_Y.$$

Riemann–Roch gives $\chi(K_Y) = 1$ and $\chi(2K_Y) = \chi(-K_Y) = 10$. The scalar part of the moving ratio is therefore

$$\frac{(x-1)^1}{(x-1)^{10}} \frac{(1-x)^{10}}{1^1} = -(1-x). \quad (7.8)$$

Since the second ideal is $\mathcal{O}_Y$, its two point-complex terms are

$$E_{12}(K_Y) = N^\vee, \quad E_{21}(-K_Y) = U \quad \text{in } K(Y^{[2]}). \quad (7.9)$$

For the first identity, apply $R\text{Hom}(-, K_Y)$ to the point-ideal sequence: the difference $[R\Gamma(K_Y)] - [R\text{Hom}(\mathcal{I}_1, K_Y)]$ is $[R\text{Hom}(\mathcal{O}_{Z_1}, K_Y)] = [N^\vee[-2]] = [N^\vee]$. For the second, apply derived sections to that sequence tensored by $-K_Y$. The even shift in the first identity contributes no minus sign in K -theory.

For any rank-two bundle F one has $e_{-u}(F^\vee) = e_u(F)$. Hence the moving point numerator $e_{x-1}(U^\vee)$ cancels the denominator $e_{1-x}(U)$, and the remaining denominator $e_{x-1}(N^\vee)$ equals $e_{1-x}(N)$. Including the high fixed Euler factor $e_x(U^\vee)$ and the insertion x^7 , the pairing for this order is

$$I = -[x^2] \int_{Y^{[2]}} \mu^2 (1-x) \frac{e_x(U^\vee)}{e_{1-x}(N)}. \quad (7.10)$$

Here is the required homogeneous expansion. Set $u_i = c_i(U)$ and $\nu_i = c_i(N)$. Then

$$e_x(U^\vee) = x^2 - u_1 x + u_2, \quad e_{1-x}(N) = (1-x)^2 + \nu_1(1-x) + \nu_2,$$

and, through degree two in the Chern classes on $Y^{[2]}$,

$$\frac{1-x}{e_{1-x}(N)} = \frac{1}{1-x} - \frac{\nu_1}{(1-x)^2} + \frac{\nu_1^2 - \nu_2}{(1-x)^3} + \text{terms of Chern degree at least three.}$$

The class μ^2 has real cohomological degree four and $Y^{[2]}$ has complex dimension four. Thus only Chern degree two in the remaining factor can contribute to the integral. Multiplication by $x^2 - u_1x + u_2$ and extraction of x^2 in that degree gives $u_2 + 2u_1\nu_1 + \nu_1^2 - \nu_2$. Equation (7.10) therefore becomes

$$I = - \int_{Y^{[2]}} \mu^2 (c_2(U) + 2c_1(U)c_1(N) + c_1(N)^2 - c_2(N)). \quad (7.11)$$

We have reduced the middle coefficient to four tautological Chern terms in (7.11). To evaluate them, consider the consecutive nested Hilbert scheme

$$\widetilde{Y}^2 = Y^{[1,2]} = \text{Bl}_{\Delta_Y}(Y^2), \quad \pi : \widetilde{Y}^2 \longrightarrow Y^2, \quad \psi : \widetilde{Y}^2 \longrightarrow Y^{[2]}.$$

The first surface coordinate records the length-one subscheme and the second the residual point. The morphism ψ is proper and generically of degree two, since either point of a reduced length-two subscheme can be selected as its length-one subscheme. The projectivization description and the tautological sequence are those of [EGL01, §1, equations (3)–(6), and Lemma 2.1], specialized to length one. Write E for the exceptional divisor and $p_i = \text{pr}_i \circ \pi$. The tautological quotient line of the projectivized diagonal ideal is $\mathcal{O}(-E)$. Consequently, for any line bundle P on Y , the tautological sequence is

$$0 \longrightarrow p_2^*P \otimes \mathcal{O}(-E) \longrightarrow \psi^*(P^{[2]}) \longrightarrow p_1^*P \longrightarrow 0. \quad (7.12)$$

It follows either from the universal point-ideal sequence, by tensoring with P and pushing along the finite universal subscheme, or directly from the quotient recording the residual point.

Use subscripts for pullbacks by p_i , and set $P = -K_Y$. The length-two incidence cycle pulls back to the sum of the two point graphs, so

$$\begin{aligned} \psi^*\mu &= \alpha_1 + \alpha_2, & \psi^*c_1(N) &= -E, & \psi^*c_2(N) &= 0, \\ \psi^*c_1(U) &= P_1 + P_2 - E, & \psi^*c_2(U) &= P_1P_2 - P_1E. \end{aligned} \quad (7.13)$$

The incidence assertion also follows by taking the codimension-two Chern character of the universal point-ideal sequence: its $\mathcal{O}(-E)$ twist does not alter the graph's fundamental codimension-two term. The bundle identities in (7.13) follow at once from (7.12), including the sign of E .

For completeness, the exceptional-divisor pushforwards needed here are

$$\pi_*E = 0, \quad \pi_*(E^2) = -[\Delta_Y]. \quad (7.14)$$

The first vanishes because E maps to a smaller-dimensional cycle. For the second, E is a $\mathbb{P}^1$ -bundle over Δ_Y and $\mathcal{O}(E)|_E = \mathcal{O}_E(-1)$, whose degree on each fiber is -1 . This proves the sign directly.

The bracket in Equation (7.11) now pulls back to

$$P_1P_2 - (3P_1 + 2P_2)E + 3E^2.$$

All terms linear in E integrate to zero by (7.14) and the projection formula. The remaining term involving P satisfies

$$\int_{Y^2} (\alpha_1 + \alpha_2)^2 P_1P_2 = 2(\alpha \cdot P)^2 = 0;$$

the two other terms in the square exceed the dimension of one surface factor. Since $\psi_*[\widetilde{Y}^2] = 2[Y^{[2]}]$, the factor of two from this cover must also be included. Thus

$$\begin{aligned} I &= -\frac{1}{2} \int_{\widetilde{Y}^2} (\alpha_1 + \alpha_2)^2 3E^2 \\ &= \frac{3}{2} \int_{\Delta_Y} (\alpha_1 + \alpha_2)^2 = \frac{3}{2} \int_Y (2\alpha)^2 = 6\alpha^2. \end{aligned}$$

The external terms pair to zero, so Equation (5.18) and $\alpha^2 \neq 0$ show that this order has universal diagonal coefficient 6.

Finally exchange the two orders. Since $L_1 = L_2 = 0$, exchanging the two summands is the same as replacing the equivariant parameter t by $-t$ before evaluation. The ordinary codimension-two output component has parameter power

$$p + 2 - v = 7 + 2 - 13 = -4.$$

This exponent is even, so the exchange preserves its pairing at $t = 1/2$. The second order therefore also has coefficient 6. Both orders and all three point-length distributions have now been accounted for, and $c_{\text{mid}} = 6 + 6 = 12$. $\square$

Proof of Proposition 7.1. Combine Equations (7.4) and (7.7) with Lemma 7.3. On the original surface X they give

$$c = (s + \tau_X)c_{\text{mid}} = 12(s + \tau_X) > 0,$$

because $s > 0$ and $\tau_X \geq 0$. $\square$

8 The integral conclusion and motivic consequences

8.1 The integral zero-cycle theorem

Both numerical branches now supply the nonzero coefficient required by Corollary 5.5. We assemble the rational vanishing and then use Roitman's theorem to remove torsion.

Proof of Theorem 1.1. For a surface of Kodaira dimension below two, the assertion follows from Bloch–Kas–Lieberman as in Proposition 2.2. For a surface of general type, pass to a smooth minimal model X_0 . If $K_{X_0}^2 \leq 8$, Proposition 6.1 gives a nonzero diagonal coefficient in Theorem 5.4 on a point blowup of X_0 . If $K_{X_0}^2 = 9$, Proposition 7.1 gives such a coefficient on X_0 itself. In either case Corollary 5.5 and Lemma 2.1 yield

$$\text{CH}_0(S)^0 \otimes_{\mathbb{Z}} \mathbb{Q} = 0.$$

Let $z \in \text{CH}_0(S)^0$. Vanishing after tensoring with $\mathbb{Q}$ means that some nonzero integer annihilates z . Roitman's torsion Theorem makes the Albanese map injective on torsion zero-cycles over $\mathbb{C}$ [Blo79, Theorem 4.2]; see also [Roi80]. But $\text{Alb}(S) = 0$, since $q(S) = 0$. Therefore $z = 0$. Thus the integral degree map is injective. Any complex point of S has degree one, so it is also surjective. $\square$

Proof of Corollary 1.2. If S is not of general type, Bloch, Kas, and Lieberman prove that the kernel of alb_S vanishes with integral coefficients [BKL76, pp. 135–136]. Surjectivity gives the claim.

Suppose S is of general type, and let S_0 be a smooth minimal model. The inequalities $K_{S_0}^2 > 0$ and $\deg c_2(S_0) > 0$, together with Noether's formula, give $1 - q(S_0) + p_g(S_0) > 0$ [PW16, Remark 1.3.1]. By Lemma 2.1, $p_g(S_0) = p_g(S) = 0$ and $q(S_0) = q(S)$, so the inequality forces $q(S) = 0$. Theorem 1.1 now gives $\text{CH}_0(S)^0 = 0$, while $\text{Alb}(S)(\mathbb{C}) = 0$. Thus alb_S is an isomorphism. $\square$

8.2 Rational Chow motives

We now pass from integral zero-cycles to rational Chow motives. Corollary 1.2 supplies the Albanese-kernel vanishing for every surface with $p_g = 0$; the additional hypothesis $q = 0$ enters only in the pure-Tate conclusions below.

Proof of Corollary 1.3. Let $T(S)$ be the kernel of the rational Albanese map

$$\mathrm{CH}_0(S)^0 \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow \mathrm{Alb}(S)(\mathbb{C}) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

Corollary 1.2 gives $T(S) = 0$. Kahn–Murre–Pedrini’s equivalence [KMP07, Corollaries 7.4.9 and 7.6.11] then gives $t_2(S) = 0$ and Kimura finite-dimensionality of $h(S)_{\mathbb{Q}}$. Their hypotheses hold over $\mathbb{C}$, an algebraically closed characteristic-zero field of infinite transcendence degree over $\mathbb{Q}$. This step does not require $q(S) = 0$.

Now assume $q(S) = 0$. Kahn–Murre–Pedrini’s surface Chow–Künneth decomposition [KMP07, Propositions 7.2.1 and 7.2.3] is

$$h(S)_{\mathbb{Q}} \simeq \mathbf{1} \oplus h_1(S) \oplus \mathbb{L}^{\oplus \rho(S)} \oplus t_2(S) \oplus h_3(S) \oplus \mathbb{L}^2,$$

where $\rho(S)$ is the Picard number. The endomorphism identifications for $h_1(S)$ and $h_3(S)$ with those of the Albanese and Picard varieties show that these two motives vanish when $q(S) = 0$. The Betti realization of the middle decomposition, with $t_2(S) = 0$, gives $\rho(S) = b_2(S)$. This gives the asserted decomposition. Tensoring it proves the assertion for all powers of S .

For a partition $\nu = 1^{a_1} \cdots n^{a_n}$ of $n \geq 1$, put $\ell(\nu) = \sum_j a_j$, $\Sigma_{\nu} = \prod_j \mathfrak{S}_{a_j}$, and $S^{(\nu)} = S^{\ell(\nu)} / \Sigma_{\nu}$. The formula of de Cataldo–Migliorini [dCM02, Theorem 6.2.1] gives

$$h(S^{[n]})_{\mathbb{Q}} \simeq \bigoplus_{\nu \vdash n} h(S^{(\nu)})_{\mathbb{Q}} \otimes \mathbb{L}^{n-\ell(\nu)}.$$

Here $h(S^{(\nu)})_{\mathbb{Q}}$ is the rational quotient motive of the possibly singular symmetric-product quotient. In the quotient formalism of [dCM02, Section 4.3, Lemma 4.3.1, and Remark 5.4.5], it is the image on $h(S^{\ell(\nu)})_{\mathbb{Q}}$ of the averaging projector

$$e_{\nu} = \frac{1}{|\Sigma_{\nu}|} \sum_{\sigma \in \Sigma_{\nu}} \Gamma_{\sigma}.$$

Since $\mathrm{Hom}(\mathbb{L}^a, \mathbb{L}^b)$ is zero for $a \neq b$ and is $\mathbb{Q}$ for $a = b$, every idempotent summand of a finite Tate sum is again a finite Tate sum. Each term in the formula is therefore pure Tate, as is $h(S^{[n]})_{\mathbb{Q}}$. The case $n = 0$ is immediate.

Finally, on a summand $\mathbb{L}^a$, the rational Chow group and even Betti cohomology each have one generator in codimension a and vanish in other codimensions; cycle class identifies these generators, and odd Betti cohomology vanishes. Compatibility of cycle class with correspondences transfers these assertions to the stated pure Tate motives. Thus the displayed cycle-class maps are isomorphisms of rational vector spaces. This uses the decomposition in rational Chow motives, not finite-dimensionality alone. $\square$

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